A

RESEARCH PROJECT REPORT

ON

"A Study on Impact of Digital Payment Adoption on Consumer Spending Behaviour Using Python and Power BI"

SUBMITTED TO

SAVITRIBAI PHULE PUNE UNIVERSITY

IN PARTIAL FULFILLMENT OF 2 YEARS FULL TIME COURSE MASTERS OF BUSINESS ADMINISTRATION (MBA)

SUBMITTED BY

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SUMMARY

In today's digital age, technology has greatly changed the way people handle financial transactions. With the increasing use of smartphones, internet services and financial technology platforms, digital payments have become a common part of everyday life. Many people now prefer using digital payment methods such as UPI, mobile wallets, debit cards, credit cards and online banking instead of carrying cash. In India, the adoption of digital payments has grown rapidly due to government initiatives promoting a cashless economy, improved internet access and the convenience offered by digital financial services. Because of these developments, consumers are gradually shifting from traditional cash payments to digital modes of payment.

As digital payment systems become more popular, they may also influence the way consumers spend their money. Consumer spending behaviour refers to how individuals decide to spend their income on goods and services. When people use cash, they can physically see the money being spent, which often makes them more careful about their expenses. However, digital payments make the process of paying much faster and easier. A transaction can be completed within seconds with just a few taps on a mobile phone. Because of this convenience, some consumers may spend more frequently or make purchases without thinking too much about their spending. At the same time, digital payment platforms also provide transaction history and spending records, which can help users keep track of their expenses and manage their finances better.

Even though digital payment systems offer many advantages such as convenience, speed and accessibility, it is important to understand how these systems affect consumer spending habits. Some researchers believe that digital payments may increase consumer spending because the payment process feels less tangible compared to paying with cash. Others argue that digital payment systems may actually help consumers manage their finances more effectively because they allow users to track their spending and analyse their financial activities. Due to these different perspectives, it becomes necessary to study how the adoption of digital payments truly impacts consumer spending behaviour.

The main purpose of this research is to examine and analyse the impact of digital payment adoption on consumer spending behaviour. The study aims to understand whether the frequent use of digital payment methods influences how consumers make purchasing decisions and manage their expenses. It will also explore factors such as convenience, ease of use, security and speed of transactions that may encourage people to adopt digital payment systems. By studying these factors, the research will attempt to identify whether consumers who regularly use digital payments show different spending patterns compared to those who prefer traditional payment methods.

To carry out this research, a quantitative research method will be used. Primary data will be collected through a structured questionnaire distributed among individuals who use digital payment methods. The respondents will mainly include students, working professionals and other consumers who perform digital transactions in their daily activities. A convenience sampling method will be used to collect responses because it allows the researcher to gather data easily and efficiently. The process of collecting responses is expected to take around three to four weeks.

The questionnaire will include questions related to the usage of digital payment platforms, frequency of digital transactions, preferred payment methods and personal spending habits. Respondents will be asked to share their opinions using a Likert scale, which helps measure their level of agreement with different statements about digital payments and spending behaviour.

After the data is collected, analytical tools such as **Python and Power BI** will be used to analyse the information. Python will be used for tasks such as data cleaning, basic statistical analysis and hypothesis testing. Power BI will be used to create visualizations and dashboards that help in understanding patterns and trends in consumer spending behaviour. Using these tools will make it easier to interpret the data and present the results in a clear and meaningful way.

This study is important because digital payments are becoming an essential part of the modern financial system. Understanding how digital payment adoption affects consumer spending behaviour can help businesses develop better marketing strategies and improve their services. Financial institutions and fintech companies can also use the insights from this research to enhance their digital payment platforms and provide better experiences for users.

Overall, this research aims to provide a better understanding of the relationship between digital payment adoption and consumer spending behaviour. By using analytical tools like Python and Power BI, the study will not only examine consumer habits but also demonstrate how data analysis can be used to generate meaningful insights in the field of business analytics. The findings of this research may help individuals, businesses and policymakers better understand how digital payments influence financial decisions in today’s digital economy.

DECLARATION

I **Anand Santosh Shelake** here-by declare that the Research Project with title "**A Study on Impact of Digital Payment Adoption on Consumer Spending Behaviour Using Python and Power BI**" is my genuine work and has been completed by me in partial fulfilment of Semester IV -MASTER OF BUSINESS ADMINISTRATION (BATCH 2024-2026) degree examination as prescribed by SINHGAD BUSINESS SCHOOL, PUNE, affiliated to SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE. This has not been submitted for any other examination and does not form the part of any other course under taken by me.

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ACKNOWLEDGEMENT

With immense pride and sense of gratitude, I take this opportunity to express my sincere regards to Dr Shriram Dawkhar Director, Sinhgad Business School, Pune. I am extremely thankful to my Research Project Mentor **Dr. Siddharth Karale** for his guidance throughout the project. I will fail in my duty if I do not thank the non-Teaching staff of the institute for their Co-operation. I would like to thank all those who helped me in making this project complete and successful.

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CERTIFICATE

This is to certify that the project report titled "**A Study on Impact of Digital Payment Adoption on Consumer Spending Behaviour using Python and Power BI** " which is being submitted herewith for the award of the degree of **Master of Business Administration (MBA)** course of **Savitribai Phule Pune University**, is the result of the Research Project completed at " **Famous Malls in -warje-pune** " by **Mr. / Ms. Shelake Anand Santosh** under our supervision and guidance.

To the best of our knowledge and belief, the work embodied in this Research Project report has not formed earlier the basis for the award of any degree or similar title of this for any other University or examination body.

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INDEX

Chapter No.	Contents	Page No.
1	**Introduction** 1.1 Overview of Research Question 1.2 Purpose /Rationale of the Project 1.3 Theoretical Framework 1.4. Significance of the Project	1 - 5
2	**Review of the Literature**	6 - 10
3	**Problem Statement & Research Hypotheses**	11 – 15
4	**Research Methodology**	16 – 19
5	**Data Analysis** 5.1 Introduction 5.2 Data Analysis Through Python 5.3 Hypothesis Testing 5.4 Data Analysis Through Power BI	20 – 49
6	**Results & Findings**	50 – 53
7	**Recommendations for further research**	54 – 55
8	**References**	56 - 57
	Annexure	58 - 63

LIST OF TABLES

Table No.	Name of Tables	Page No.
5.1	Data Description	22
5.2	Count of Lickert Scale Questions	40
5.3	Observed frequency table for Hypothesis 1	41
5.4	Expected frequency table for Hypothesis 1	41
5.5	Chi-Square value table for Hypothesis 1	42
5.6	Observed frequency table for Hypothesis 2	43
5.7	Expected frequency table for Hypothesis 2	43
5.8	Chi-Square value table for Hypothesis 2	44
5.9	Observed frequency table for Hypothesis 3	45
5.10	Expected frequency table for Hypothesis 3	45

5.11	Chi-Square value table for Hypothesis 3	**46**
5.12	Summary of Hypothesis Testing Result	**53**

LIST OF IMAGES

Image No.	Name of Image	Page No.
5.1	Install and Import Libraries	**23**
5.2	Load and Preview Dataset	**24**
5.3	Data Cleaning	**25**
5.4	Age Analysis	**26**
5.5	Gender Analysis	**26**
5.6	Income Analysis	**27**
5.7	Usage Frequency Analysis	**28**
5.8	Purpose of Payment Analysis	**29**
5.9	Reason for Use	**29**
5.10	Spending Behaviour Analysis	**30**
5.11	Cashback Impact Analysis	**31**
5.12	Impulse Purchase Analysis	**32**
5.13	Gender Vs Spending More	**33**
5.14	Usage Frequency vs Purchase Increase	**34**
5.15	Income Vs Spending Behaviour	**35**
5.16	Gender Vs Usage Frequency	**36**
5.17	Convert Likert Scale into Numbers	**37**
5.18	Mean Value	**37**
5.19	Correlation Analysis	**38**
5.20	Price Segmentation Analysis	**39**
5.21	Compare Spending Analysis	**39**
5.22	Get Data in Power BI	**47**
5.23	Digital Payment Adoption & Consumer Spending Behaviour Analysis Dashboard	**48**

CHAPTER I

INTRODUCTION

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INTRODUCTION

1.1 OVERVIEW OF THE RESEARCH QUESTION:

In recent years, digital technology has significantly transformed the way financial transactions are conducted. Traditional cash-based transactions are gradually being replaced by digital payment systems such as mobile wallets, debit cards, credit cards, internet banking and Unified Payments Interface (UPI). These digital payment methods enable consumers to complete transactions quickly, securely and conveniently through smartphones or online platforms. As a result, digital payments have become an essential part of the modern financial ecosystem.

India has experienced remarkable growth in digital payment adoption, especially in the last decade. Government initiatives promoting a cashless economy, the expansion of digital infrastructure and the increasing use of smartphones and internet services have contributed to the widespread acceptance of digital payment platforms. Applications such as mobile wallets and UPI-based services allow consumers to transfer money, pay bills and make purchases instantly. These innovations have made financial transactions more accessible and efficient for individuals and businesses alike.

As digital payment systems become more popular, they may also influence consumer spending behaviour. Consumer spending behaviour refers to the patterns and decisions individuals make when purchasing goods and services. The convenience and speed of digital payment systems may encourage consumers to make more frequent purchases or increase the amount they spend. Additionally, promotional incentives such as cashback offers, reward points and discounts provided by digital payment platforms may further motivate consumers to adopt these payment methods.

The increasing use of digital payments raises an important research question: **Does the adoption of digital payment systems influence consumer spending behaviour?** Understanding this relationship is important because digital payments not only change the method of transaction but may also affect how consumers manage and spend their money.

This research aims to analyse the impact of digital payment adoption on consumer spending behaviour by using data analytics techniques. The study utilizes tools such as **Python for data analysis and Power BI for data visualization** to examine patterns and trends in consumer payment and spending behaviour. By applying analytical techniques, the research attempts to

provide a deeper understanding of how digital payment usage affects consumer purchasing decisions.

1.2 PURPOSE OF THE PROJECT:

The primary purpose of this research project is to examine the relationship between digital payment adoption and consumer spending behaviour using analytical methods. As digital payment systems continue to grow in popularity, it becomes increasingly important to understand how these systems influence consumer financial habits.

Digital payment platforms provide several advantages compared to traditional cash transactions. These include convenience, speed, security and accessibility. Consumers can complete transactions instantly without the need to carry physical cash or visit banks and ATMs. This ease of transaction may influence consumer purchasing behaviour by making spending more frequent and effortless.

Another important factor that encourages digital payment adoption is the availability of promotional benefits such as cashback offers, discounts, loyalty rewards and special deals. These incentives often motivate consumers to choose digital payment methods over cash transactions. As a result, digital payment platforms may indirectly influence the purchasing decisions and spending patterns of consumers.

The rationale behind conducting this study is to analyse these behavioural changes using a data-driven approach. By applying analytical tools such as Python and Power BI, the research can identify patterns and trends in consumer spending behaviour related to digital payment usage. Python helps in performing statistical analysis, data cleaning and hypothesis testing, while Power BI helps in visualizing the results through interactive dashboards and charts.

This research is also relevant for businesses and financial institutions that rely on digital payment systems. Understanding consumer spending patterns can help companies design better marketing strategies, improve digital payment services and create attractive promotional campaigns. Additionally, the study provides valuable insights for policymakers who aim to promote digital financial inclusion and strengthen the digital economy.

Therefore, the purpose of this project is not only to study consumer behaviour but also to use modern analytical tools to provide meaningful insights about the impact of digital payment adoption on spending patterns.

1.3 THEORETICAL FRAMEWORK:

The theoretical framework of this study is based on the relationship between technology adoption and consumer behaviour. Several theories in technology management and consumer psychology help explain why individuals adopt new technologies and how these technologies influence their decision-making processes.

One of the most relevant theories for this research is the Technology Acceptance Model (TAM). According to this model, individuals are more likely to adopt new technology if they perceive it as useful and easy to use. In the context of digital payments, consumers are more likely to use digital payment systems if they find them convenient, reliable and secure. When consumers perceive these systems as beneficial, the likelihood of adoption increases.

Another important concept related to this study is Consumer Behaviour Theory, which explains how individuals make purchasing decisions based on psychological, economic and social factors. When digital payment options simplify the payment process, consumers may experience less hesitation while making purchases. This may increase the frequency of transactions and influence spending patterns.

Additionally, the concept of Behavioural Economics suggests that consumers do not always make completely rational financial decisions. When using digital payment methods, consumers do not physically see or feel the exchange of money as they do when using cash. This can create what is known as the "cashless effect," where consumers may spend more because the payment process feels less tangible.

In this research framework, digital payment adoption acts as the independent variable**,** while consumer spending behaviour acts as the dependent variable. Factors such as convenience, security, ease of use and promotional incentives are analysed to determine how they influence consumer spending patterns.

The analytical component of this research further strengthens the theoretical framework. By using Python for statistical analysis and Power BI for visualization, the study can explore relationships between variables and test hypotheses effectively. This analytical approach helps transform theoretical concepts into measurable insights.

1.4 SIGNIFICANCE OF THE PROJECT:

The significance of this research lies in its ability to provide valuable insights into the changing financial behaviour of consumers in the digital age. As digital payment technologies continue to

expand, understanding their impact on consumer spending behaviour becomes increasingly important for businesses, financial institutions, and policymakers.

For businesses and retailers, this study provides insights into how digital payment systems influence customer purchasing behaviour. By understanding these patterns, businesses can develop better marketing strategies, offer targeted promotions and improve customer engagement. Companies can also use these insights to optimize their payment systems and enhance the overall customer experience.

For financial institutions and fintech companies, the findings of this research can help improve digital payment platforms by focusing on factors that influence consumer adoption. Enhancing security, improving user experience and offering attractive incentives can encourage more consumers to adopt digital payment systems.

The research is also significant from a policy perspective. Governments around the world, including India, are promoting digital financial systems to increase transparency, reduce cash dependency and improve financial inclusion. Understanding the impact of digital payment adoption on consumer behaviour can help policymakers design more effective strategies to support the growth of digital economies.

From an academic perspective, this research contributes to the field of Business Analytics by demonstrating how analytical tools such as Python and Power BI can be used to study consumer financial behaviour. The integration of data analytics with consumer behaviour research provides a modern and practical approach to understanding real-world business problems.

In conclusion, digital payment systems are transforming the way consumers conduct financial transactions. By analysing the impact of digital payment adoption on consumer spending behaviour through analytical tools and data-driven methods, this research aims to provide meaningful insights into the evolving relationship between technology and consumer financial decisions.

CHAPTER II

REVIEW OF THE LITERATURE

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REVIEW OF THE LITERATURE

1. **Impact of digital payment methods on consumer trust and spending behaviour: evidence from India** (Sheikh Gowher Rasool and Tahir Ahmad Wani) (February 2, 2026)

   The current study attempts to examine the perception of safety for consumers associated with two payment options, namely Unified Payment Interface and card payments, across high (laptop) and low involvement (mobile charger) purchase, utilising behavioural economics perspectives. Purposive and snowball sampling was used and a total of 383 respondents were approached to participate in the survey out of which 367 agreed to participate however 322 responses were considered for study. Participants were divided into groups based on payment options (UPI and card) and transaction involvement (purchase of laptop and mobile charger). Results indicate no significant difference in safety perception among participants between UPI and card payments, suggesting consumers weigh benefits and risks similarly for both methods. Perceived safety positively influences trust, aligning with behavioural economics theories emphasising heuristics and biases in decision making. Trust, in turn, positively impacts consumer spending, mediated by convenience. However, loss aversion negatively moderates the relationship between trust and spending growth, indicating that loss-averse individuals may resist increasing spending regardless of trust in electronic payments. These findings underscore the importance of cultivating trust and convenience to encourage payment method adoption in emerging countries like India which leads to increased consumer spending, while considering individual differences in loss aversion.

2. **Effects of Mobile Payment Systems on Consumer Spending and Saving behaviour** (Dave Michael)

   The rapid expansion of mobile payment systems has significantly transformed consumer financial behaviour by changing how individuals conduct transactions, manage expenses, and save money. In both developing and emerging economies, mobile payment platforms have become increasingly popular due to their convenience, accessibility, and integration with everyday economic activities. This study examines the effects of mobile payment systems on consumer spending and saving behaviour. Using a quantitative analytical approach, the study investigates how mobile payment adoption influences consumption patterns, spending frequency, impulse purchases, and saving habits. The findings suggest that mobile payment usage is associated with increased transaction frequency and consumer spending, while its impact on saving behaviour depends on user awareness, income level, and financial literacy. The study contributes to the

literature on digital finance and consumer economics by providing empirical evidence on the behavioural implications of mobile payment systems. The results offer important insights for policymakers, financial service providers, and consumer protection agencies seeking to promote responsible digital payment usage and financial well-being.

3. **Impact Of Digital Payment Systems on Consumer Spending Habits.** (Kanwal, Jaismeen; Singh, Jagpreet)

   Digital payment systems (UPI, mobile wallets, contactless cards) have rapidly reshaped how consumers transact, potentially altering spending frequency, ticket size, budgeting discipline, and impulse purchases. This study investigates how adoption of digital payments affects consumer spending habits in urban and semi-urban India. Using a structured questionnaire (N = 840) and a difference-in-differences design comparing self-reported monthly discretionary spending before and after adoption with non-adopters as a comparison group we examine mechanisms including convenience, perceived security, rewards/cashback, and financial tracking features. Results indicate a statistically significant increase in spending frequency and average monthly discretionary spend (+9–15% across age cohorts), partially mediated by convenience and rewards, and moderated negatively by high financial literacy. While digital payments improve record-keeping and budget visibility, this does not fully offset convenience-driven overspending for younger users. The paper concludes with policy and managerial implications for responsible product design (soft limits, default budgeting nudges), consumer education, and more nuanced loyalty programs.

4. **How digital payment methods shape consumer spending behaviour in Finland** (Le, Duong 2024)

   This thesis explores the impact of digital payment methods on consumer behaviour in Finland, focusing on how they affect spending habits and budgeting. The study looks at how different age groups use digital payments and how this influences their ability to manage their finances. A survey was conducted among Finnish consumers aged 18 and older to understand how they use digital payment methods and manage their finances. The results show that all consumers use digital payments very often, however some struggle with budgeting. Based on these findings, the study suggests adding features like budgeting tools to digital payment systems to help users keep better control over their spending. The research highlights how digital payment systems are influencing consumer behaviour and offers suggestions for improving financial management through these technologies.

5. **Impact of digital payments on financial inclusion and consumer financial behaviour in India.**

   Research conducted by **Geeta Sharma (2019)** focused on the impact of digital payments on financial inclusion and consumer financial behaviour in India. The study concluded that digital payment systems improve financial accessibility and encourage consumers to shift from cash-based transactions to electronic transactions.

6. **Digital Payment Adoption in India**

   Shampy Kamboj and Yogesh K. Dwivedi (2020) conducted research on digital payment adoption in India. Their study revealed that demographic factors, perceived security and ease of use significantly affect consumer willingness to adopt digital payment systems. The research also highlighted the importance of digital literacy in increasing the adoption of electronic payment methods.

7. **Consumer Perceptions of Digital Payment Platforms**

   Neeraj Arora and K. K. Sahney (2018) examined consumer perceptions of digital payment platforms. The researchers found that promotional offers such as cashback, discounts and reward points play an important role in encouraging consumers to adopt digital payment systems. These incentives often lead to increased spending and higher transaction frequency.

8. **Mobile Payment Adoption and User Satisfaction**

   Ozlem Turel, **Alexander Serenko** and **Nick Bontis (2007)** focused on mobile payment adoption and user satisfaction. Their findings indicated that convenience, security and trust in technology significantly influence users' intention to adopt mobile payment services. The study also emphasized that positive user experiences increase the frequency of digital payment usage.

9. **Digital payment technology and consumer behaviour – Saving, spending patterns: Are saving and spending patterns a concern?** (Ramasundaram Gurusamy and Aiswarya B) (May 24, 2023)

   Do not save what is left after spending but spend what is left after saving. *Warren Buffet* Our spending habits are shaped by our upbringing, our values, ethics, our view of life, security, and one's expectations of the future. However, the mode of payment also has an impact on the spending patterns of an individual. It is clear that the introduction of digital payments has affected spending patterns and subsequently savings habits. People generally aren't getting better at saving

through technology they are getting better at spending. The case examines and highlights the behaviour of individuals who use cash and cashless digital payments in conjunction with their respective spending and saving patterns.

10. **Digital Payment and Customer behaviour (How Mobile Payment Systems Influence the Spending Habits of International Students in Finland)** Hamal, Milan (2025)

    This study was aimed at analysing the impact of mobile payment systems on the spending, budgeting, and general financial management behaviour of international students in Finland. The thesis discusses the convenience and potential costs of popular digital payment networks such as Mobile Pay, Google Pay, Apple Pay and Siirto. The research was conducted using quantitative interviews with international students in order to provide a comprehensive picture of the effects of financial behaviours in a cashless environment.

    The theoretical framework was based on Unified Theory of Acceptance and Use of Technology (UTAUT), and its main factors are performance expectancy, effort expectancy, social influence, and facilitating conditions. Financial knowledge, trust and cultural adaptation were also considered to be influential in the study. The results show that mobile payments enhance financial tracking with digital receipts and transaction histories but also enhance the chances of unplanned expenses. Most students, particularly those who were previously exposed to the digital finance, showed high confidence in the digital payment infrastructure of Finland and managed to perform well in the cashless environment. It was also found that mobile payments improved the socialization of students.

CHAPTER III

PROBLEM STATEMENT & RESEARCH HYPOTHESES

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PROBLEM STATEMENT & RESEARCH HYPOTHESES

3.1 PROBLEM STATEMENT:

The rapid growth of digital technology has significantly transformed the financial ecosystem across the world. In recent years, digital payment systems such as mobile wallets, debit cards, credit cards, internet banking and Unified Payments Interface (UPI) have become increasingly popular among consumers. These payment systems allow individuals to complete financial transactions quickly, conveniently and securely through digital platforms. In countries like India, the adoption of digital payments has increased rapidly due to the expansion of smartphone usage, improved internet connectivity and government initiatives promoting a cashless economy.

Digital payment platforms provide several advantages over traditional cash transactions. They offer convenience, speed, ease of use and additional benefits such as cashback offers, discounts and reward points. These features may encourage consumers to prefer digital payment methods over cash transactions. As a result, digital payment systems are not only changing the way people make transactions but may also influence how consumers spend their money.

However, despite the growing use of digital payment systems, there is still limited understanding of how these systems affect consumer spending behaviour. Consumer spending behaviour refers to the patterns and decisions individuals follow when purchasing goods and services. The convenience of digital payment methods may encourage consumers to make more frequent purchases or increase their overall spending. On the other hand, some consumers may continue to follow cautious spending habits regardless of the payment method used.

Understanding this relationship is important for businesses, financial institutions and policymakers. Businesses need to understand how payment methods influence customer purchasing decisions in order to design effective marketing strategies and promotional campaigns. Financial institutions and fintech companies also need insights into consumer payment preferences to improve digital payment services and enhance user experience.

Furthermore, with the increasing availability of transaction data, analytical tools such as **Python** and **Power BI** can be used to analyse patterns in digital payment usage and consumer spending behaviour. These tools help researchers identify trends, relationships and insights from large datasets.

Therefore, the main problem addressed in this research is to examine whether the adoption of digital payment systems has a significant impact on consumer spending behaviour. The study aims to analyse this relationship using analytical techniques and data visualization tools to provide meaningful insights into consumer financial behaviour in the digital economy.

3.2 RESEARCH HYPOTHESES:

A research hypothesis is a statement that predicts the relationship between variables in a study. In this research, digital payment adoption is considered the **independent variable**, while consumer spending behaviour is considered the **dependent variable**.

The following hypotheses are formulated for this study:

Null Hypothesis (H_0):

There is **no significant impact** of digital payment adoption on consumer spending behaviour.

Alternative Hypothesis (H_1):

There is a **significant impact** of digital payment adoption on consumer spending behaviour.

SPECIFIC HYPOTHESIS:

Hypothesis 1 -

H_{01}: Transactions_Easier has no significant impact on **Spend_More_than_Cash**

H_{11}: Transactions_Easier has a significant impact on **Spend_More_than_Cash**

Hypothesis 2

H_{02}: Purchase_Frequency_Increase has no significant impact on **Spend_More_than_Cash**

H_{12}: Purchase_Frequency_Increase has a significant impact on **Spend_More_than_Cash**

Hypothesis 3

H_{03}: Cashback_Encourages_Spending has no significant impact on **Spend_More_than_Cash**

H_{13}: Cashback_Encourages_Spending has a significant impact on **Spend_More_than_Cash**

These hypotheses will be tested through data analysis using statistical and analytical techniques. By analysing transaction data and consumer payment patterns, the study aims to determine whether digital payment adoption influences consumer spending frequency, purchasing patterns or overall expenditure.

The results of this analysis will help provide a clearer understanding of the relationship between digital payment usage and consumer spending behaviour, contributing valuable insights for businesses, financial institutions and policymakers.

3.3 VARIABLES OF THE STUDY:

In any research study, variables play an important role in understanding the relationship between different factors. A variable refers to any measurable factor or characteristic that can change or vary within a study. In this research, variables are used to analyse how the adoption of digital payment systems influences consumer spending behaviour.

The present study mainly focuses on two types of variables: Independent Variables and Dependent Variables. These variables help in identifying the relationship between digital payment adoption and the spending behaviour of consumers.

Independent Variable:

The **independent variable** is the factor that influences or causes changes in another variable. In this study, the independent variable is **Digital Payment Adoption**. It refers to the use of digital payment methods such as UPI, mobile wallets, debit cards, credit cards and internet banking for making financial transactions. Factors such as convenience, ease of use, security and promotional offers influence the adoption of digital payment systems.

Independent Variables (Digital Payment Adoption)

- Transactions_Easier
- Purchase_Frequency_Increase
- Cashback_Encourages_Spending

Dependent Variable:

The **dependent variable** is the outcome that is affected by the independent variable. In this research, the dependent variable is **Consumer Spending Behaviour**. It refers to how consumers

spend money on goods and services, including the frequency of purchases, the amount spent, and changes in purchasing patterns.

Dependent Variable (Spending Behaviour)

- Spend_More_than_Cash

Relationship Between Variables:

The main objective of this research is to analyse the relationship between digital payment adoption and consumer spending behaviour. It is assumed that the increased use of digital payment methods may influence how frequently consumers make purchases and how much they spend.

By using analytical tools such as **Python** for data analysis and **Power BI** for data visualization, the study will examine patterns and trends in consumer payment behaviour. These tools will help in identifying whether there is a significant relationship between digital payment adoption and consumer spending behaviour.

Understanding this relationship will provide valuable insights into how technological advancements in financial systems are influencing consumer decision-making and purchasing habits in the modern digital economy.

CHAPTER IV

RESEARCH METHODOLOGY

CHAPTER IV

RESEARCH METHODOLOGY

4.1 RESEARCH METHODOLOGY:

Research methodology refers to the systematic process used to collect, analyse and interpret data in order to answer the research question. In this study titled **"An Analytical Study on Impact of Digital Payment Adoption on Consumer Spending Behaviour Using Python and Power BI"** a structured research methodology is used to examine the relationship between digital payment usage and consumer spending patterns.

4.2 DATA COLLECTION:

The data for this study is collected using **primary and secondary sources**.

Primary Data:

Primary data is collected through a structured questionnaire distributed to consumers who regularly use digital payment methods such as UPI, mobile wallets, debit cards and internet banking. The questionnaire includes questions related to digital payment usage, spending frequency and consumer preferences.

Secondary Data:

Secondary data is collected from research articles, journals, websites and reports related to digital payments and consumer behaviour. These sources help in understanding existing research and provide background information for the study.

4.3 SAMPLE SIZE:

For the purpose of this research, a **sample size 153 respondents** are considered. These respondents represent consumers who actively use digital payment platforms. The sample size is selected to obtain meaningful insights into consumer spending behaviour and digital payment usage patterns.

4.4 SAMPLING PROCEDURE:

The study uses Convenience Sampling, which is a non-probability sampling technique. In convenience sampling, respondents are selected based on their accessibility and willingness to participate in the survey.

This sampling method is commonly used in academic research because it allows researchers to collect data quickly and efficiently within limited time and resources. The respondents are selected from easily accessible groups such as students, working professionals and individuals who frequently use digital payment systems.

Although convenience sampling may not fully represent the entire population, it is suitable for exploratory and analytical studies such as this research.

4.5 SAMPLING UNIT:

The sampling unit refers to the individual elements from which the sample is drawn. In this study, the sampling unit consists of individual consumers who use digital payment methods for financial transactions.

These consumers may include:

- Students
- Working professionals
- Business owners
- Online shoppers
- Individuals who frequently use mobile payment applications

The study focuses on individuals who regularly use digital payment platforms, as their experiences and behaviours provide useful insights into the impact of digital payment adoption on spending behaviour.

4.6 STATISTICAL TOOLS USED:

To analyse the collected data, analytical and statistical tools are used. The study applies **data analysis and visualization techniques** using:

- **Chi-Square Test** is used for to test the Hypothesis. This Analysis was carried out using Microsoft Excel.
- **Python** for data cleaning, statistical analysis and hypothesis testing.

- **Power BI** for creating dashboards, charts and visual reports to understand spending patterns.

In addition, basic statistical techniques such as **percentage analysis, correlation analysis and graphical representation** may be used to interpret the results effectively.

4.7 DATA ANALYSIS METHOD:

The collected data was first organised and cleaned using Excel, Python. After that it was grouped into categorised to make it suitable for Chi-square test.

Each Hypothesis was tested separately using a structured approach:

1. Data Classification into categories
2. Preparation of contingency table
3. Calculation of expected values
4. Application of Chi-Square formula
5. Comparison with critical value
6. Conclusion (accept or reject hypothesis)

4.8 STUDY LIMITATIONS:

Although the research aims to provide meaningful insights, the study has certain limitations:

- The study is limited to a specific number of respondents and may not represent the entire population.
- This study is geographically limited to a specific area of Warje, Pune city in Maharashtra, India. The data has been collected only from respondents residing in and around this region. Therefore, the findings of the study may reflect the digital payment behaviour and spending patterns of consumers within this particular area only.
- The research mainly focuses on consumers who use digital payment systems and the behaviour of non-digital users is not included.
- The responses collected through questionnaires depend on the honesty and understanding of the respondents.
- Time constraints and limited resources may restrict the scope of data collection and analysis.

CHAPTER V

DATA ANALYSIS

CHAPTER V

DATA ANALYSIS

5.1 INTRODUCTION

This chapter presents the analysis and interpretation of data collected from 153 respondents through a structured questionnaire. The objective of this analysis is to understand the impact of digital payment adoption on consumer spending behaviour. The collected data has been analysed using Python for statistical calculations and Power BI for visualization purposes.

The analysis includes demographic profiling, usage patterns of digital payments, consumer spending behaviour, perception analysis and hypothesis testing.

5.2 DATA ANALYSIS THROUGH PYTHON

Data Description:

No.	Column Name	Description
1.	Age	Age group of the respondent (e.g., 21–30, 31–40)
2.	Gender	Gender of respondent (Male/Female)
3.	Occupation	Profession (Student, Employee, Business, etc.)
4.	Monthly_Income	Income range of respondent
5.	Internet_Usage	Frequency of internet usage (Daily, Weekly, etc.)
6.	Digital_Payment_Method	Preferred method (UPI, Cards, Wallets, etc.)
7.	Usage_Frequency	How often digital payments are used
8.	Purpose_of_Payment	Reason (shopping, bills, recharge, etc.)
9.	Years_Using_Digital_Payments	Experience with digital payments

10.	Reason_for_Using	Why they use it (convenience, cashback, etc.)
11.	Transactions_Easier	Whether digital payments make transactions easier
12.	Purchase_Frequency_Increase	Increase in purchase frequency
13.	Spend_More_than_Cash	Spending more compared to cash
14.	Cashback_Encourages_Spending	Effect of cashback offers
15.	Online_Shopping_Increase	Increase in online shopping
16.	Impulse_Purchase_Frequency	Frequency of impulse buying
17.	Purchase_Type	Type of purchases (planned/unplanned)
18.	Convenience_Rating	Rating of ease of use
19.	Security_Perception	How safe users feel
20.	Experienced_Fraud	Whether fraud was experienced
21.	Trust_for_Large_Transactions	Trust level for big payments
22.	Helps_Track_Spending	Helps in budgeting/tracking
23.	Supports_Cashless_Economy	Opinion on digital economy
24.	Overall_Satisfaction	Overall satisfaction with digital payments

Table No. 5.1: Data Description

Python Codes for Data Analysis:

INSTALL AND IMPORT LIBRARIES

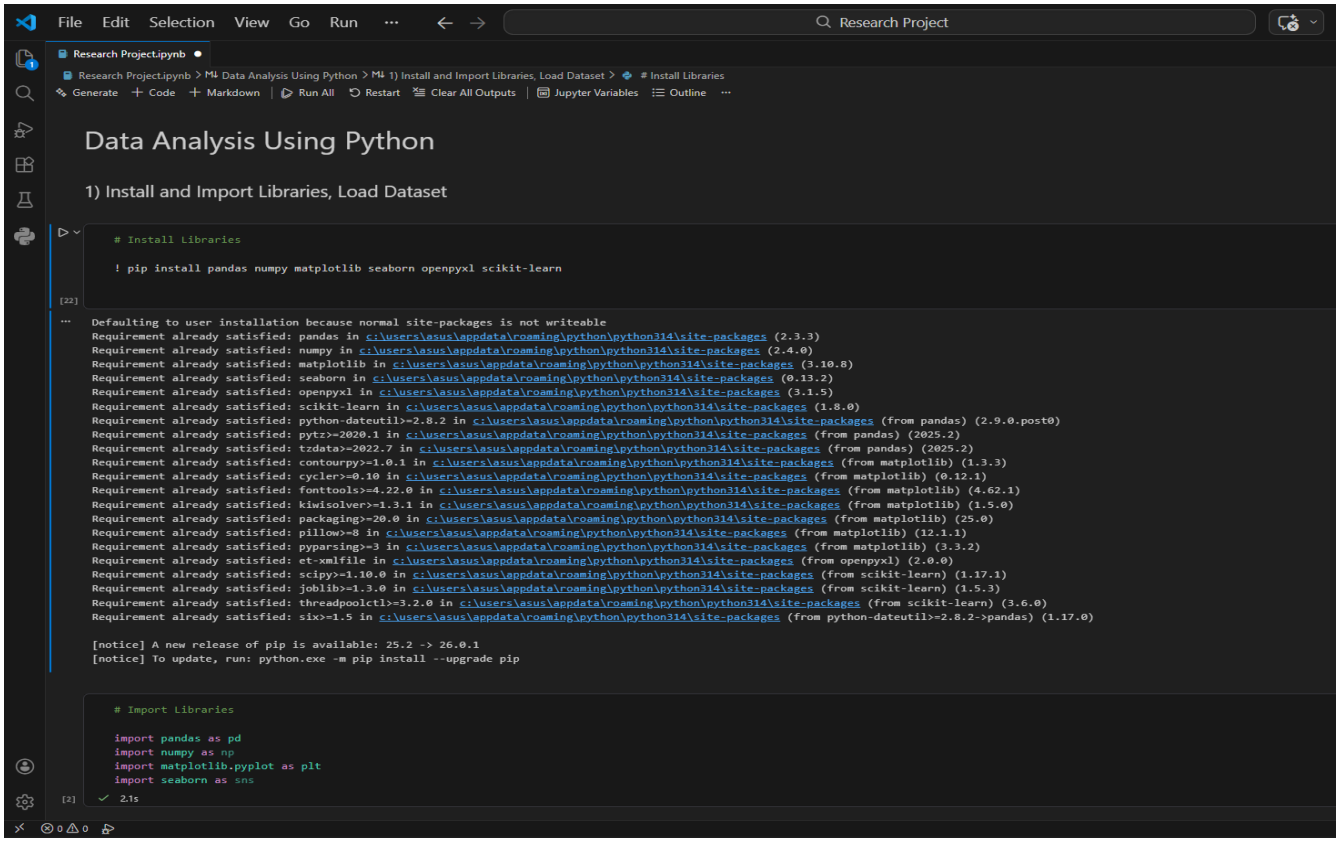
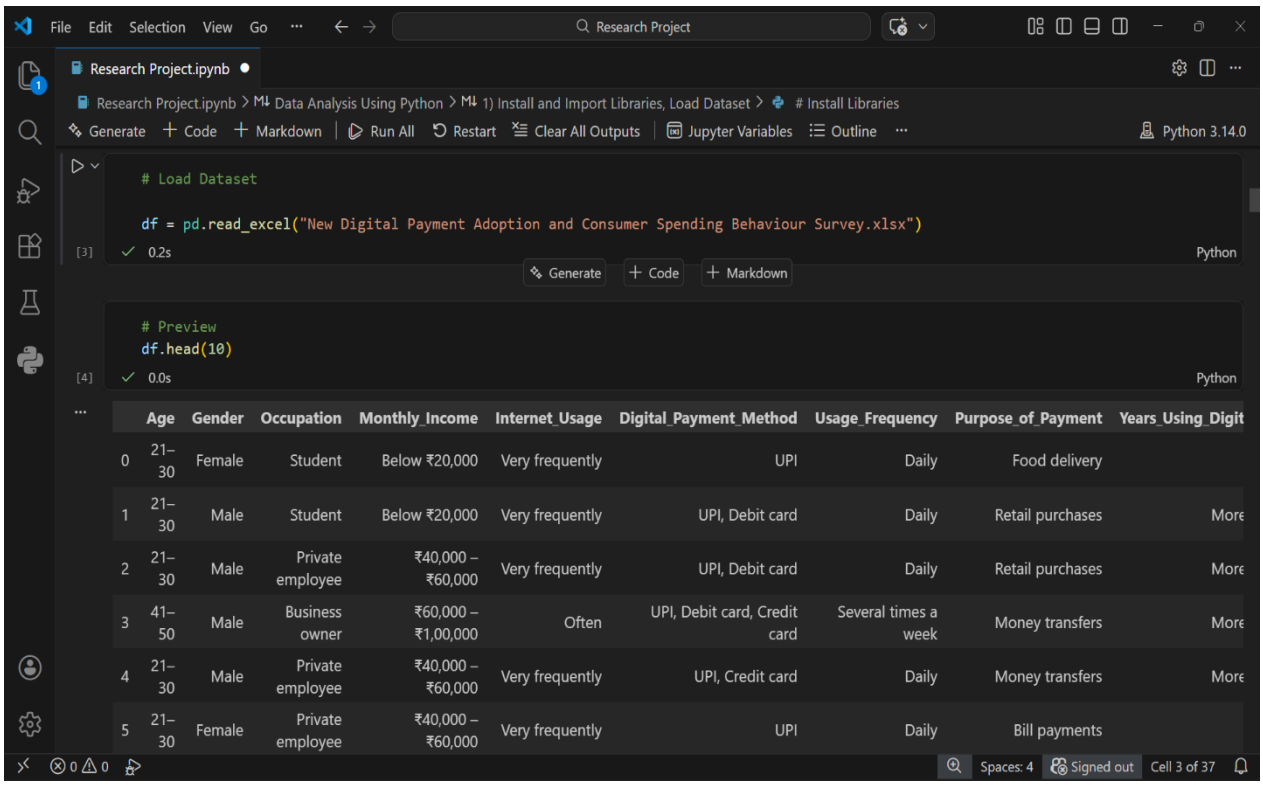


IMAGE NO. 5.1 Install and Import Libraries

INTERPRETATION:

This step represents the foundation of the data analysis process. Required Python libraries such as pandas, numpy, matplotlib and seaborn are installed and imported to perform data manipulation, statistical analysis and visualization. This ensures that all necessary tools are available for efficient and accurate analysis.

LOAD AND PREVIEW DATASET



	Age	Gender	Occupation	Monthly_Income	Internet_Usage	Digital_Payment_Method	Usage_Frequency	Purpose_of_Payment	Years_Using_Digit
0	21–30	Female	Student	Below ₹20,000	Very frequently	UPI	Daily	Food delivery	
1	21–30	Male	Student	Below ₹20,000	Very frequently	UPI, Debit card	Daily	Retail purchases	More
2	21–30	Male	Private employee	₹40,000 – ₹60,000	Very frequently	UPI, Debit card	Daily	Retail purchases	More
3	41–50	Male	Business owner	₹60,000 – ₹1,00,000	Often	UPI, Debit card, Credit card	Several times a week	Money transfers	More
4	21–30	Male	Private employee	₹40,000 – ₹60,000	Very frequently	UPI, Credit card	Daily	Money transfers	More
5	21–30	Female	Private employee	₹40,000 – ₹60,000	Very frequently	UPI	Daily	Bill payments	

IMAGE NO. 5.2 Load and Preview Dataset

INTERPRETATION:

The dataset is successfully loaded into the Python environment and a preview of the data is displayed. This helps in understanding the structure, types of variables and sample responses. It also ensures that the dataset has been imported correctly before proceeding with further analysis.

DATA CLEANING

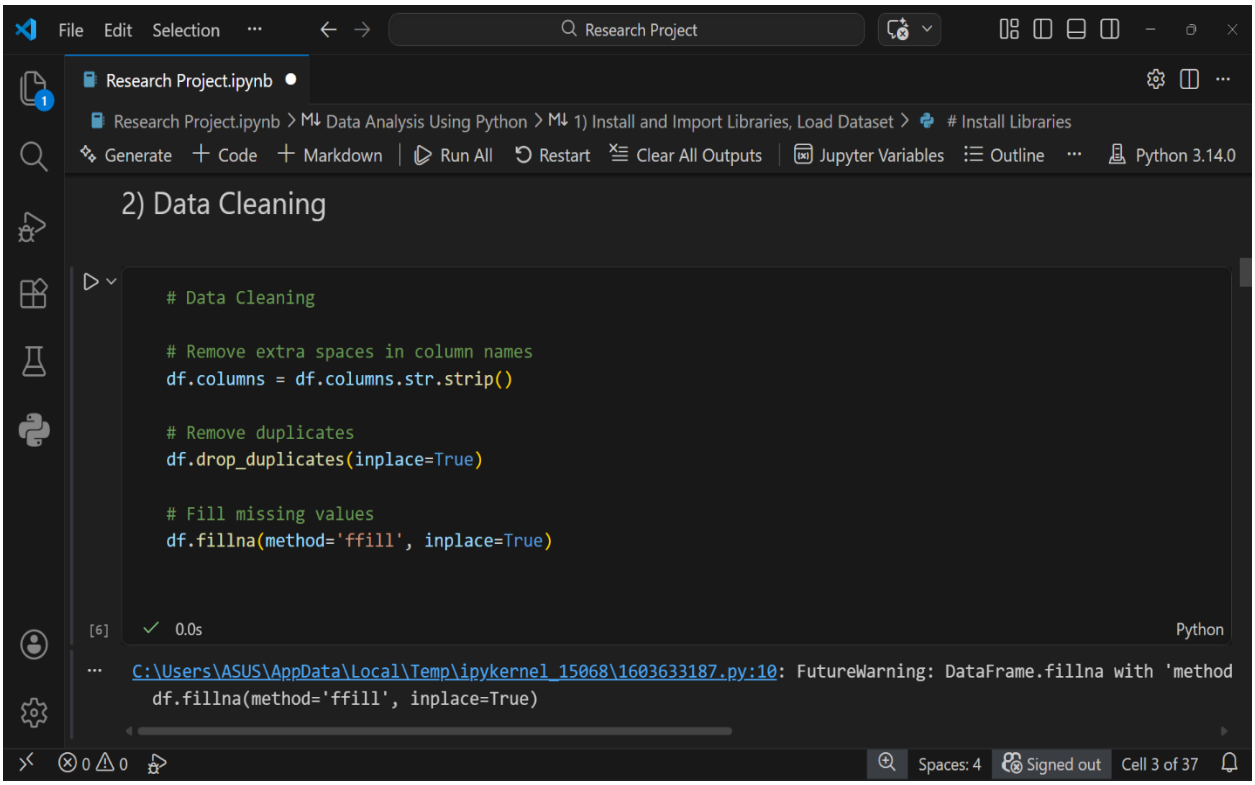


IMAGE NO. 5.3 Data Cleaning

INTERPRETATION:

Data cleaning is performed to handle missing values, remove inconsistencies, and standardize the dataset. This step is essential to improve the reliability and validity of the analysis, ensuring that the results obtained are accurate and meaningful.

AGE ANALYSIS

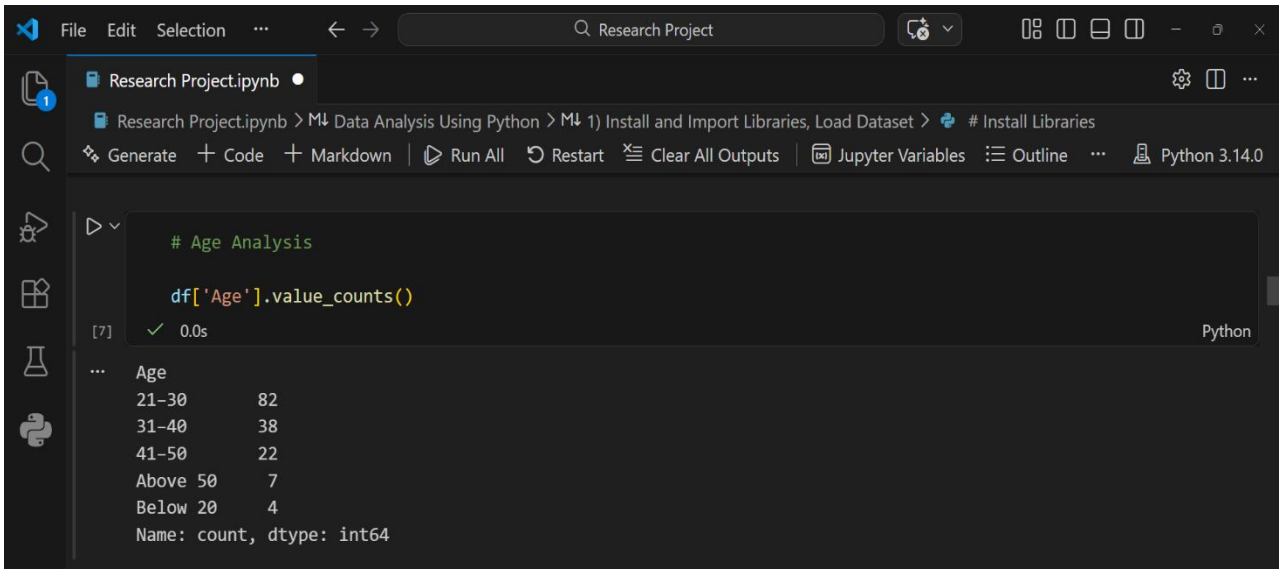


IMAGE NO. 5.4 Age Analysis

INTERPRETATION:

The age distribution clearly shows that the majority of respondents belong to the 21–30 age group, with the highest number of responses coming from this category. The 31–40 age group has moderate participation, while very few respondents fall into the 41–50, above 50 and below 20 categories. This indicates that younger individuals are more actively using digital payment systems, mainly due to their familiarity with technology and digital platforms.

GENDER ANALYSIS

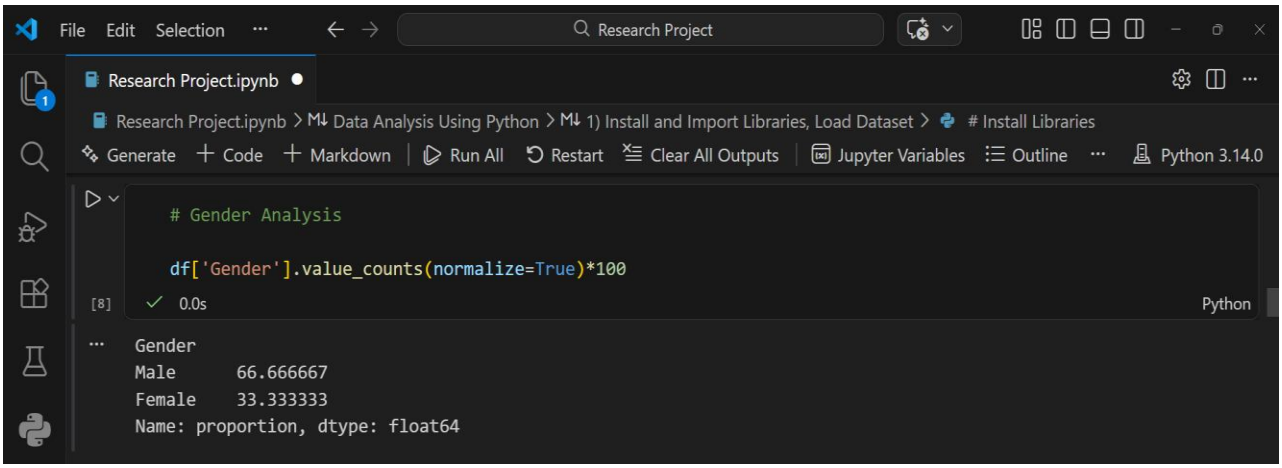


IMAGE NO. 5.5 Gender Analysis

INTERPRETATION:

Out of the total 153 respondents, 102 respondents (66.67%) are male and 51 respondents (33.33%) are female. This shows that male participation is higher in the study. However, both genders are actively using digital payments, which means adoption is not limited to a single gender group.

INCOME ANALYSIS

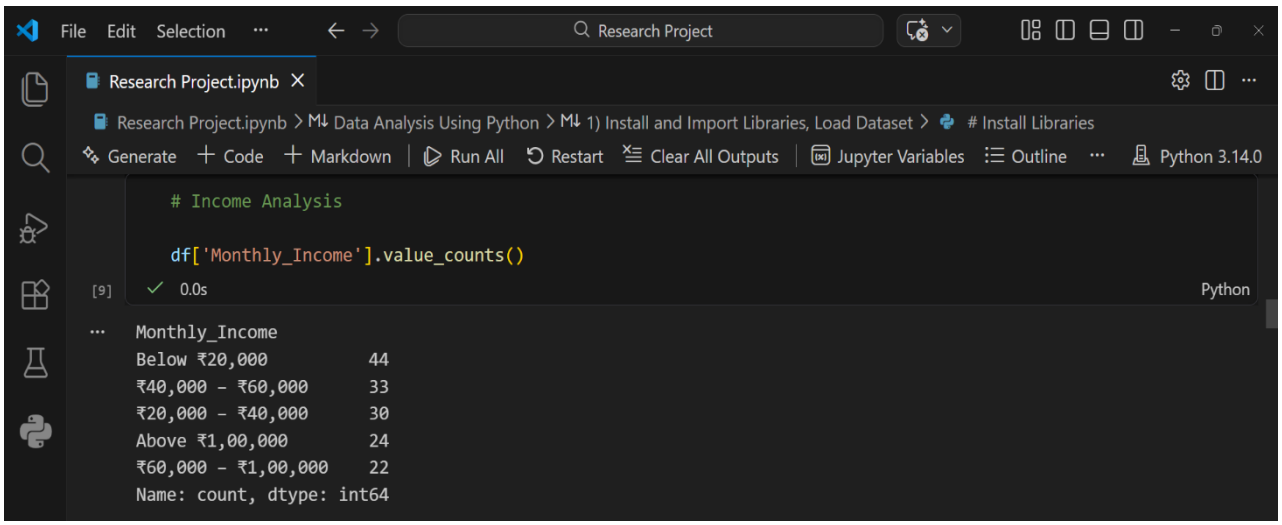


IMAGE NO. 5.6 Income Analysis

INTERPRETATION:

The income distribution shows that a large number of respondents fall into the middle-income category, with fewer respondents in the high-income and low-income groups. This suggests that digital payment systems are widely used across different income levels and are not restricted to high-income individuals only.

USAGE FREQUENCY ANALYSIS

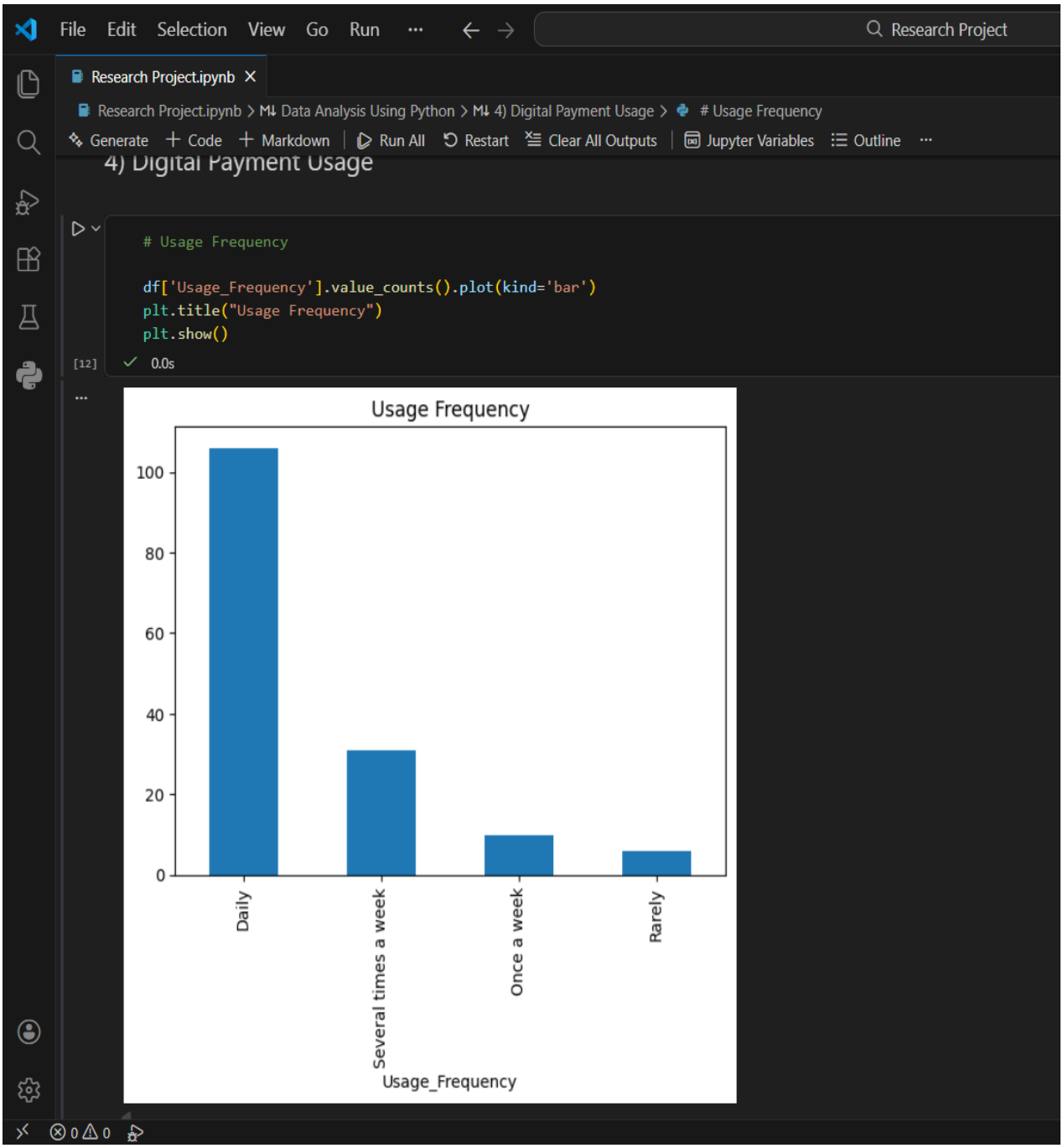


IMAGE NO. 5.7 Usage Frequency Analysis

INTERPRETATION:

The analysis shows that more than 100 respondents use digital payments daily, making it the most common usage pattern. Around 30–40 respondents use it several times a week, while only a small number of users use it once a week or rarely. This clearly indicates that digital payments have become a regular part of daily transactions.

PURPOSE OF PAYMENT ANALYSIS

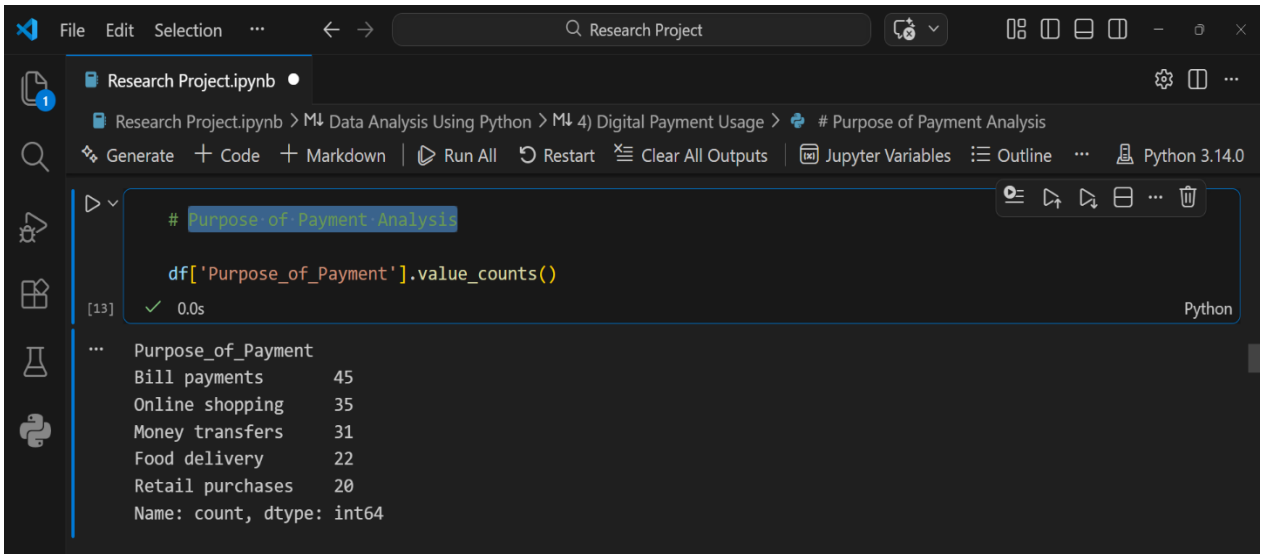


IMAGE NO. 5.8 Purpose of Payment Analysis

INTERPRETATION:

The purpose-wise distribution shows that 29.41% of respondents use digital payments for bill payments, making it the most common purpose. This is followed by 22.88% for online shopping and 20.26% for money transfers. The remaining respondents use digital payments for food delivery and retail purchases. This shows that digital payments are widely used for multiple daily activities.

REASON FOR USE

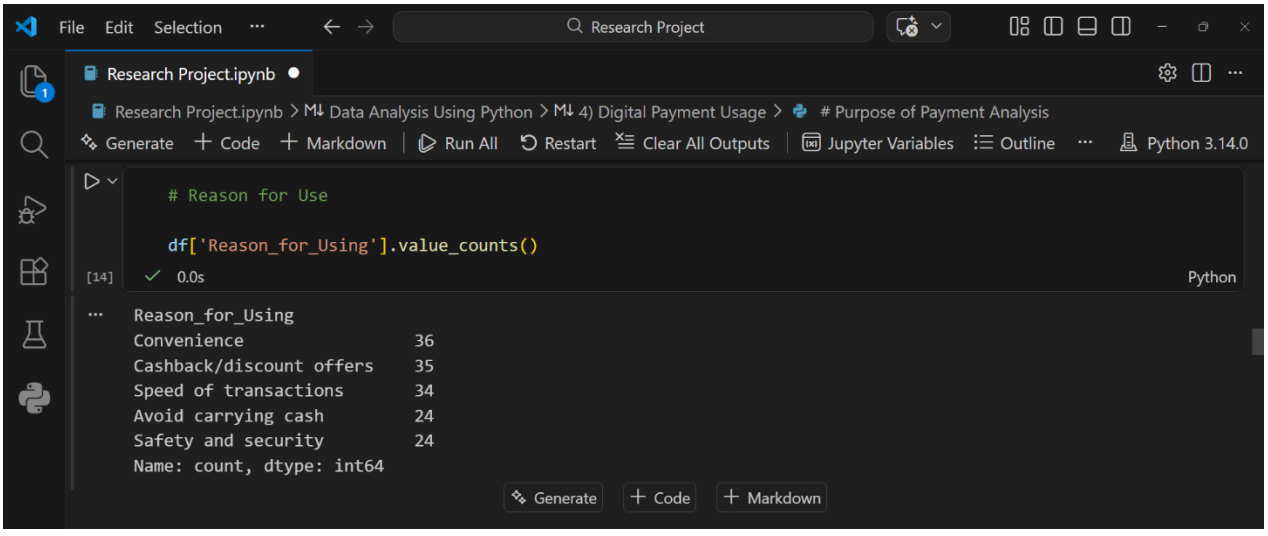


IMAGE NO. 5.9 Reason for Use

INTERPRETATION:

The primary reasons for using digital payments include convenience, speed and cashback offers. These factors act as strong motivators for consumers and significantly contribute to the adoption of digital payment systems.

SPENDING BEHAVIOUR ANALYSIS

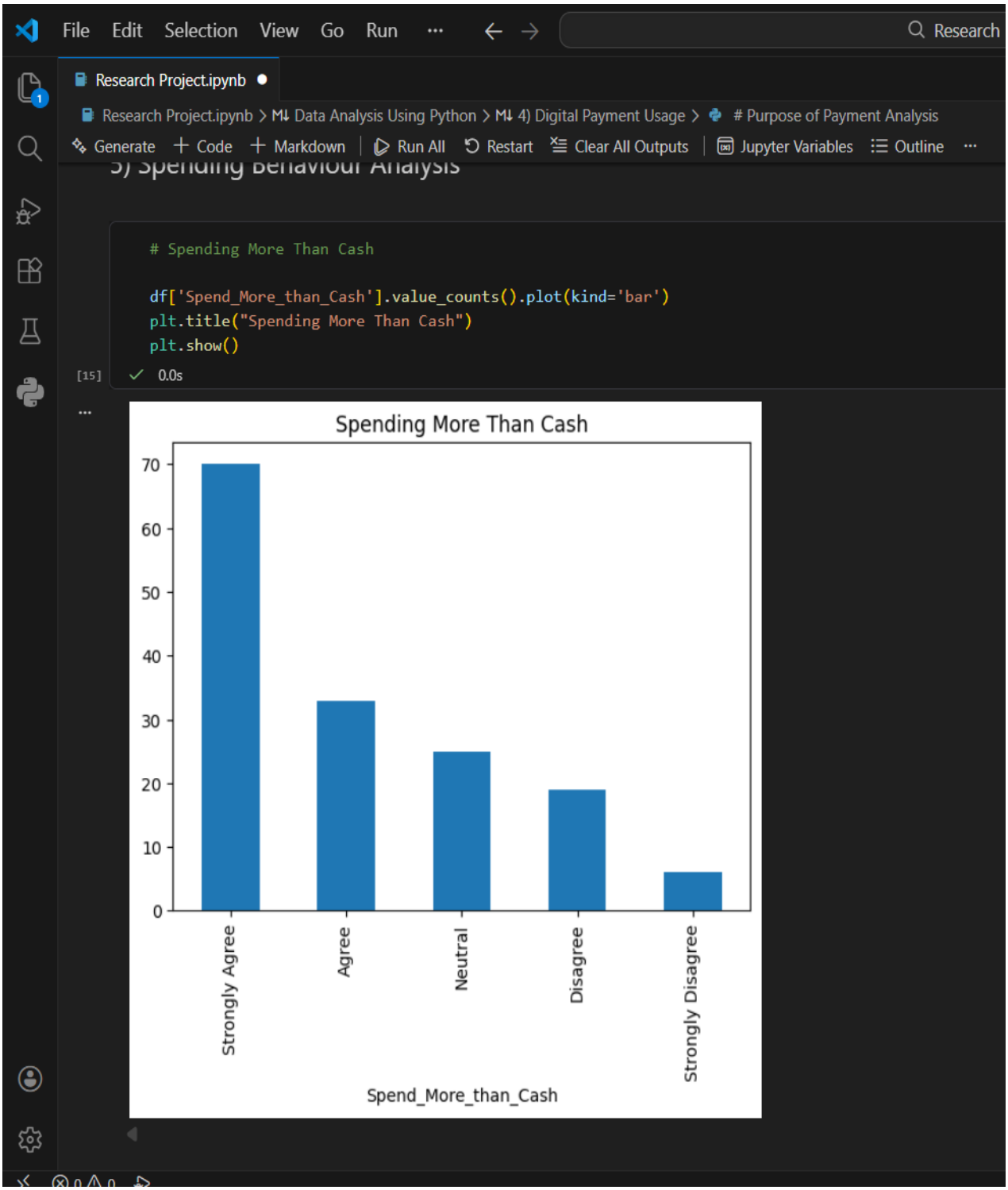


IMAGE NO. 5.10 Spending Behaviour Analysis

INTERPRETATION:

The responses indicate that a majority of users fall under “Agree” and “Strongly Agree” categories, suggesting that digital payments increase their spending behaviour. Very few respondents selected neutral or disagree options. This means that most users feel they tend to spend more when using digital payment methods compared to cash.

CASHBACK IMPACT ANALYSIS

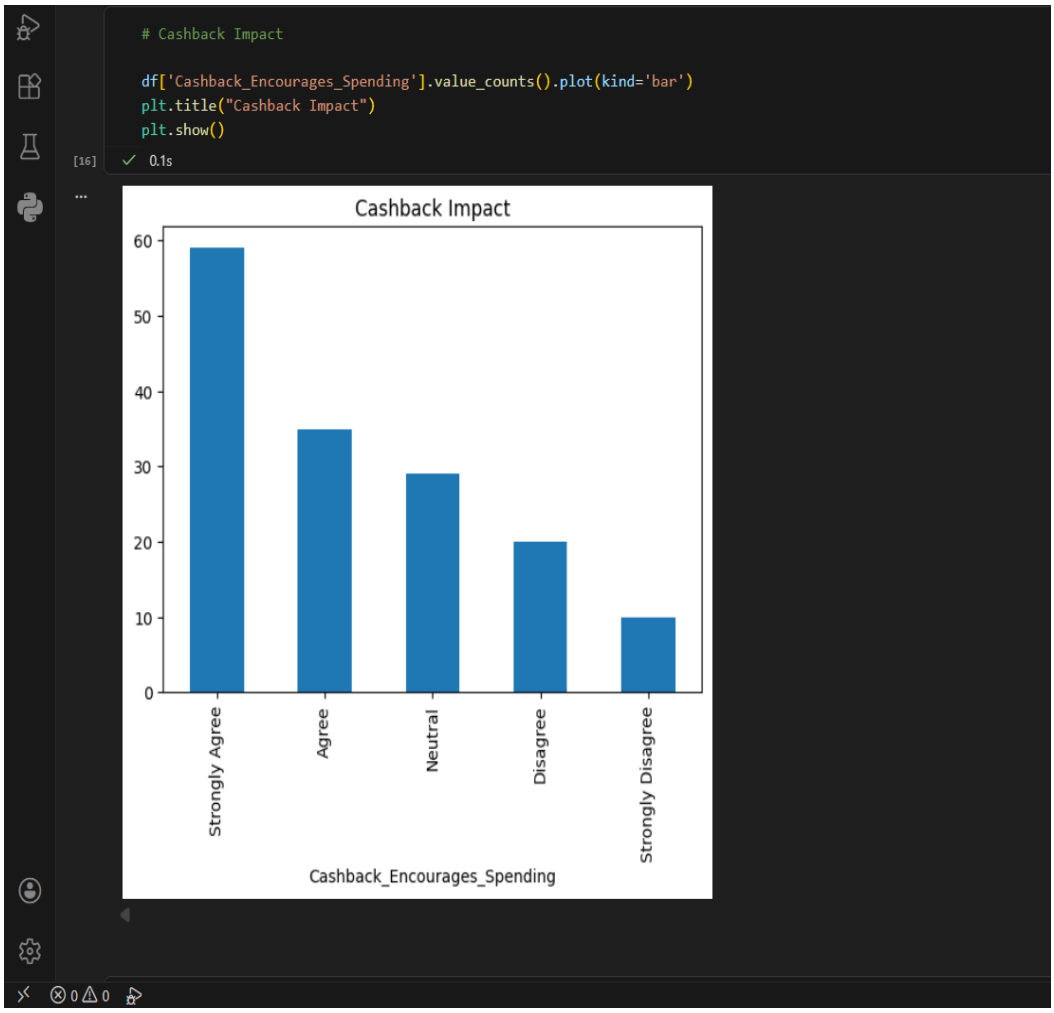


IMAGE NO. 5.11 Cashback Impact Analysis

INTERPRETATION:

A large number of respondents have selected “Agree” and “Strongly Agree” for cashback influence, showing that offers and discounts play a major role in encouraging spending. Only a small number of users are neutral or disagree, indicating that cashback is a strong motivating factor.

IMPULSE PURCHASE ANALYSIS

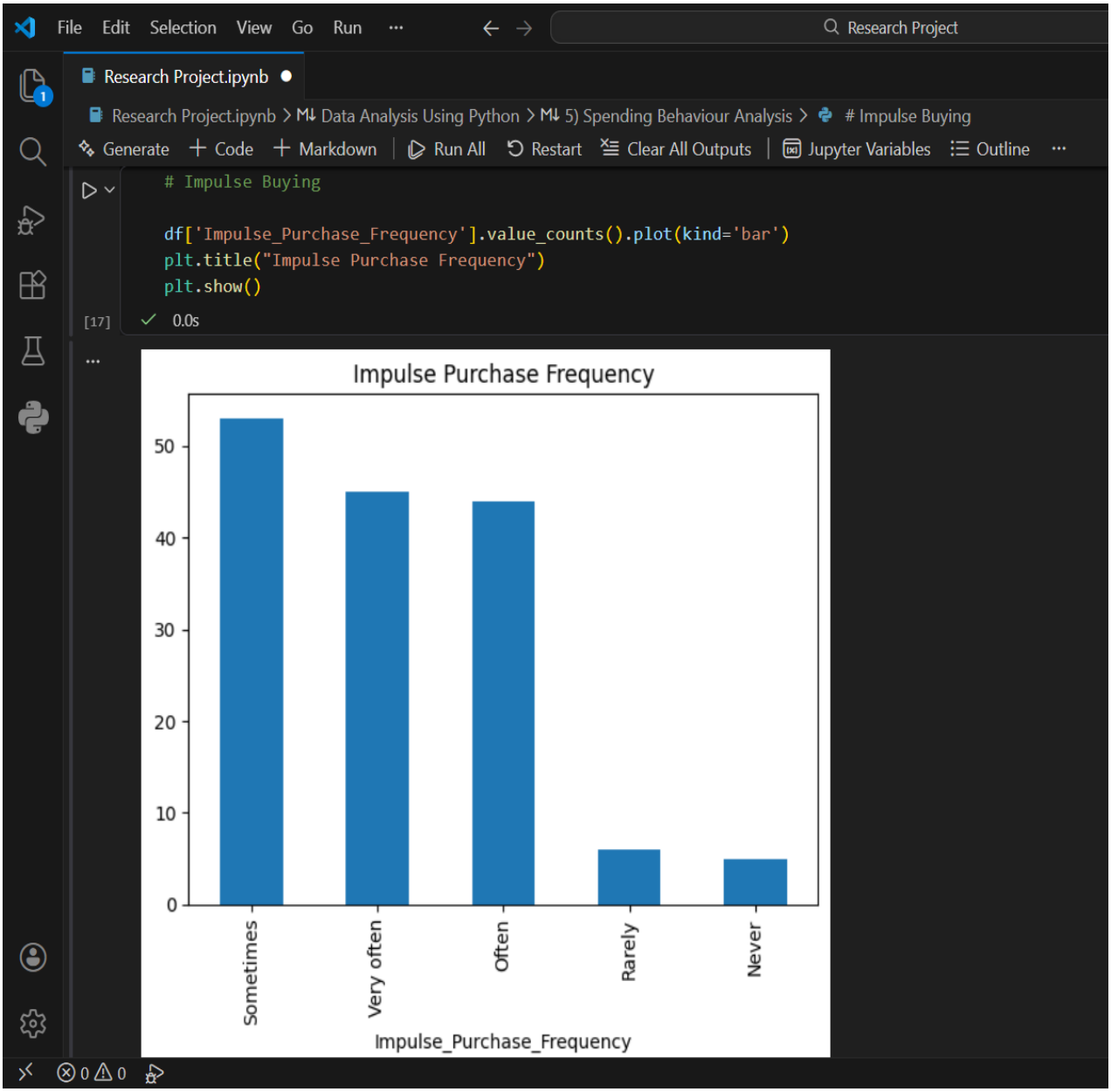


IMAGE NO. 5.12 Impulse Purchase Analysis

INTERPRETATION:

The findings suggest that digital payments increase the likelihood of impulse purchases. The ease and speed of transactions reduce hesitation, leading consumers to make unplanned purchases more frequently.

GENDER VS SPENDING MORE

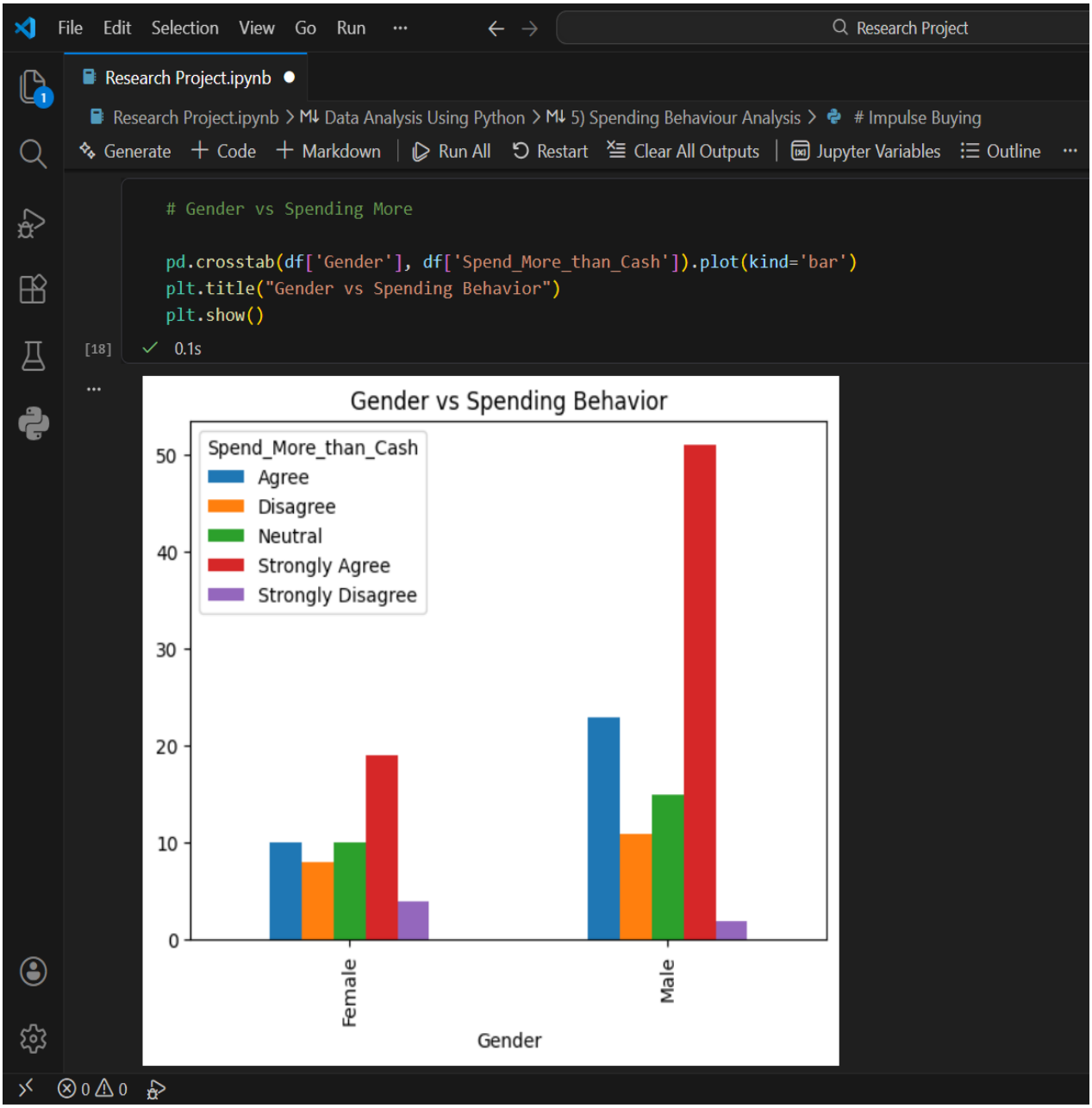


IMAGE NO. 5.13 Gender Vs Spending More

INTERPRETATION:

The comparison between gender and spending behaviour shows that both male and female respondents tend to spend more when using digital payments. However, since the number of male respondents (102) is higher, their overall contribution to spending appears greater. Still, the trend remains similar across both genders.

USAGE FREQUENCY VS PURCHASE INCREASE

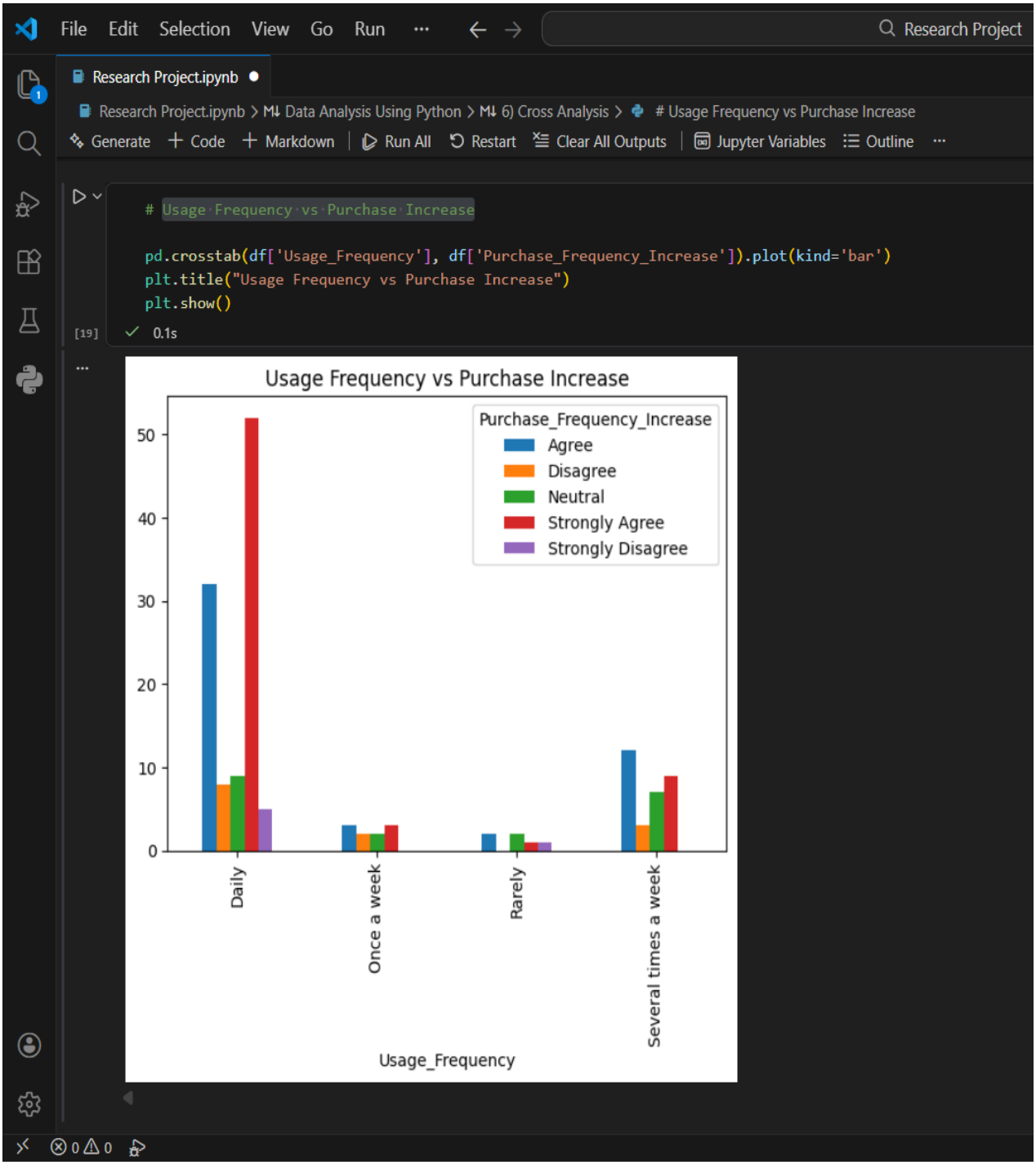


IMAGE NO. 5.14 Usage Frequency vs Purchase Increase

INTERPRETATION:

Respondents who use digital payments daily show the highest level of increased purchase frequency. As the usage frequency decreases (weekly or rarely), the level of spending also reduces. This indicates a direct relationship between how often digital payments are used and how much consumers spend.

INCOME VS SPENDING BEHAVIOUR

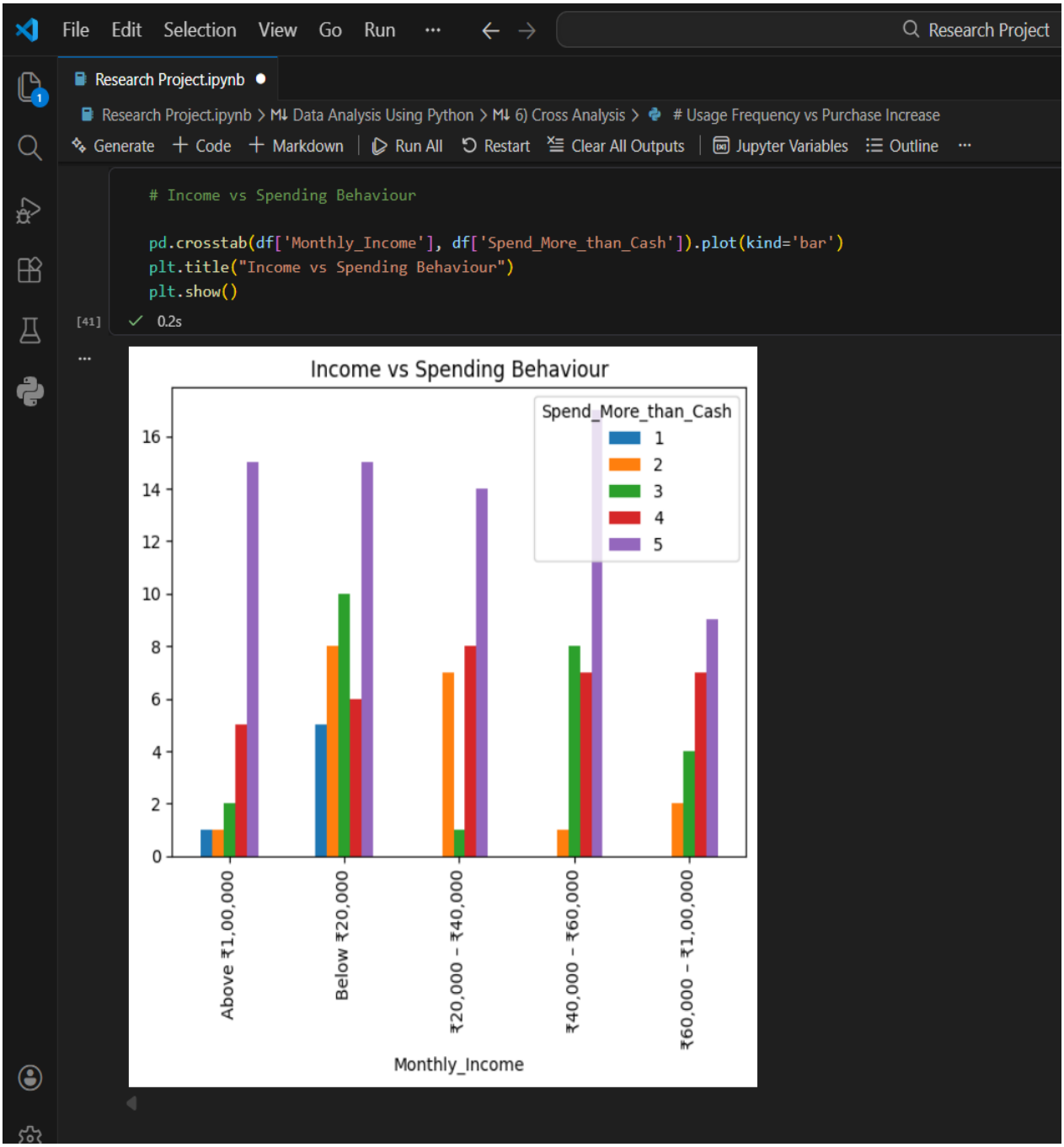


IMAGE NO. 5.15 Income Vs Spending Behaviour

INTERPRETATION:

The analysis shows that respondents from all income groups tend to spend more using digital payments. However, middle- and higher-income groups show slightly higher spending levels compared to lower income groups. This suggests that income has some influence, but digital payments impact all groups.

GENDER VS USAGE FREQUENCY

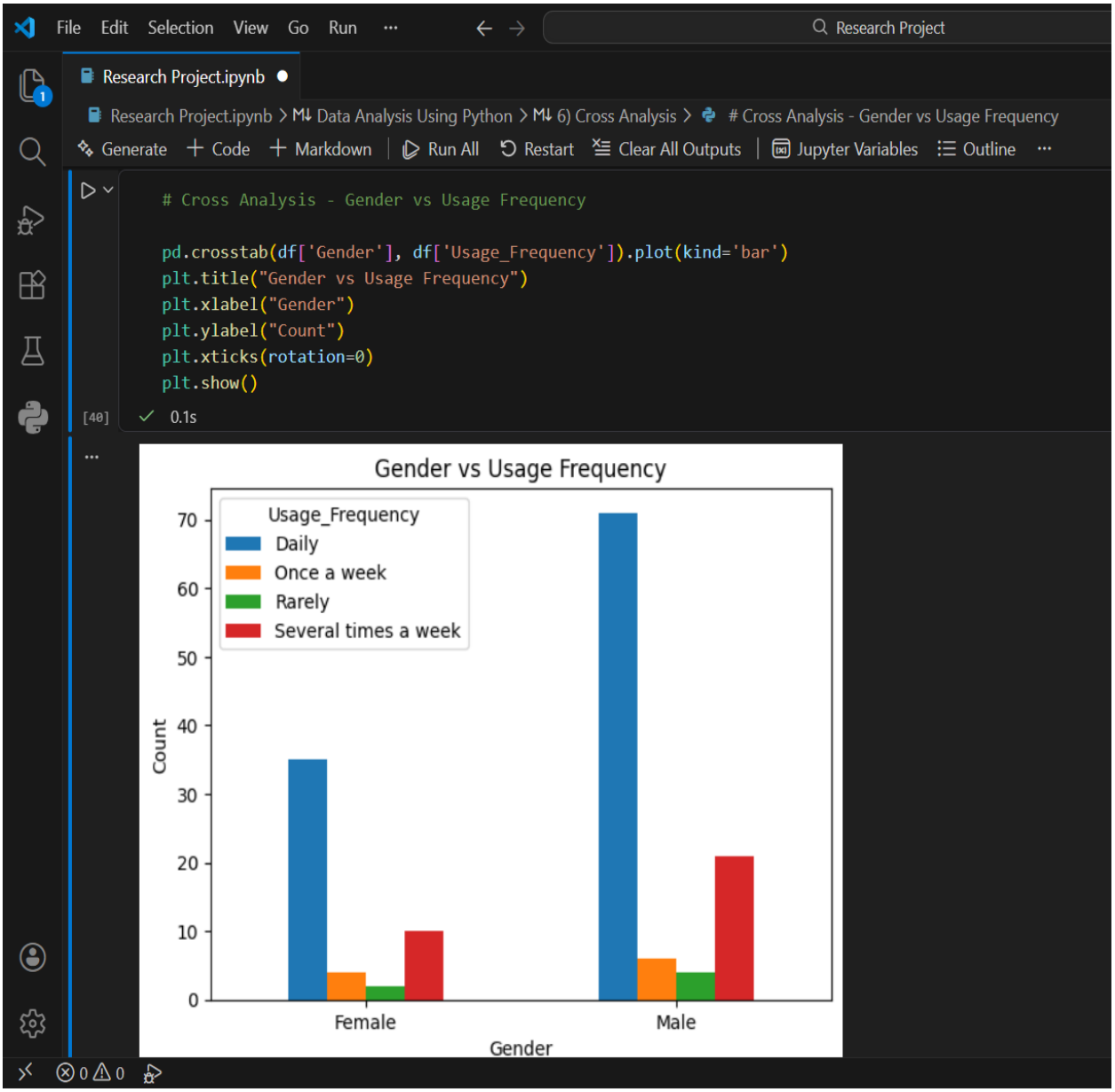


IMAGE NO. 5.16 Gender Vs Usage Frequency

INTERPRETATION:

The analysis indicates that both genders actively use digital payment systems with similar frequency. This reflects the widespread acceptance and usage of digital payments across different demographic groups.

CONVERT LIKERT SCALE INTO NUMBERS

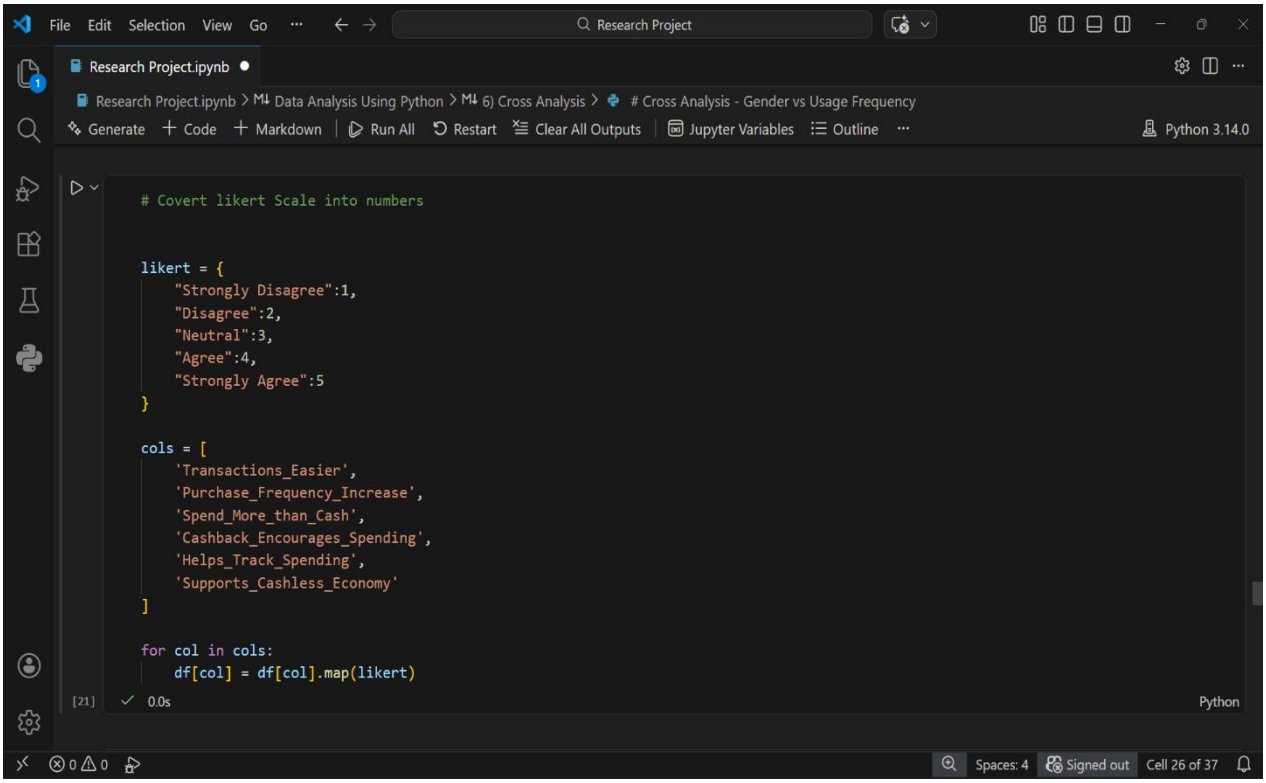


IMAGE NO. 5.17 Convert Likert Scale into Numbers

INTERPRETATION:

The conversion of qualitative responses into numerical values allows for statistical analysis such as mean, correlation and hypothesis testing. This step is essential for transforming subjective opinions into measurable data.

MEAN VALUE

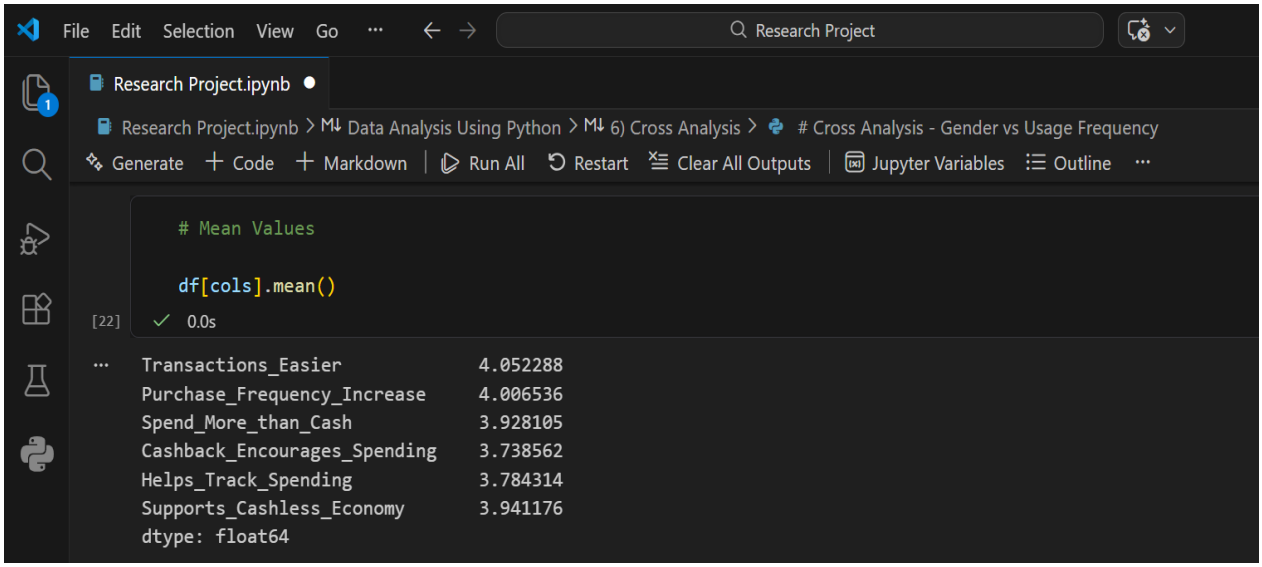


IMAGE NO. 5.18 Mean Value

INTERPRETATION: The mean values of key variables are above 3.5, with spending behaviour at 3.93 and convenience at 4.05. This indicates that most respondents agree that

digital payments are convenient and lead to increased spending. These values clearly show a positive perception among users.

CORRELATION ANALYSIS

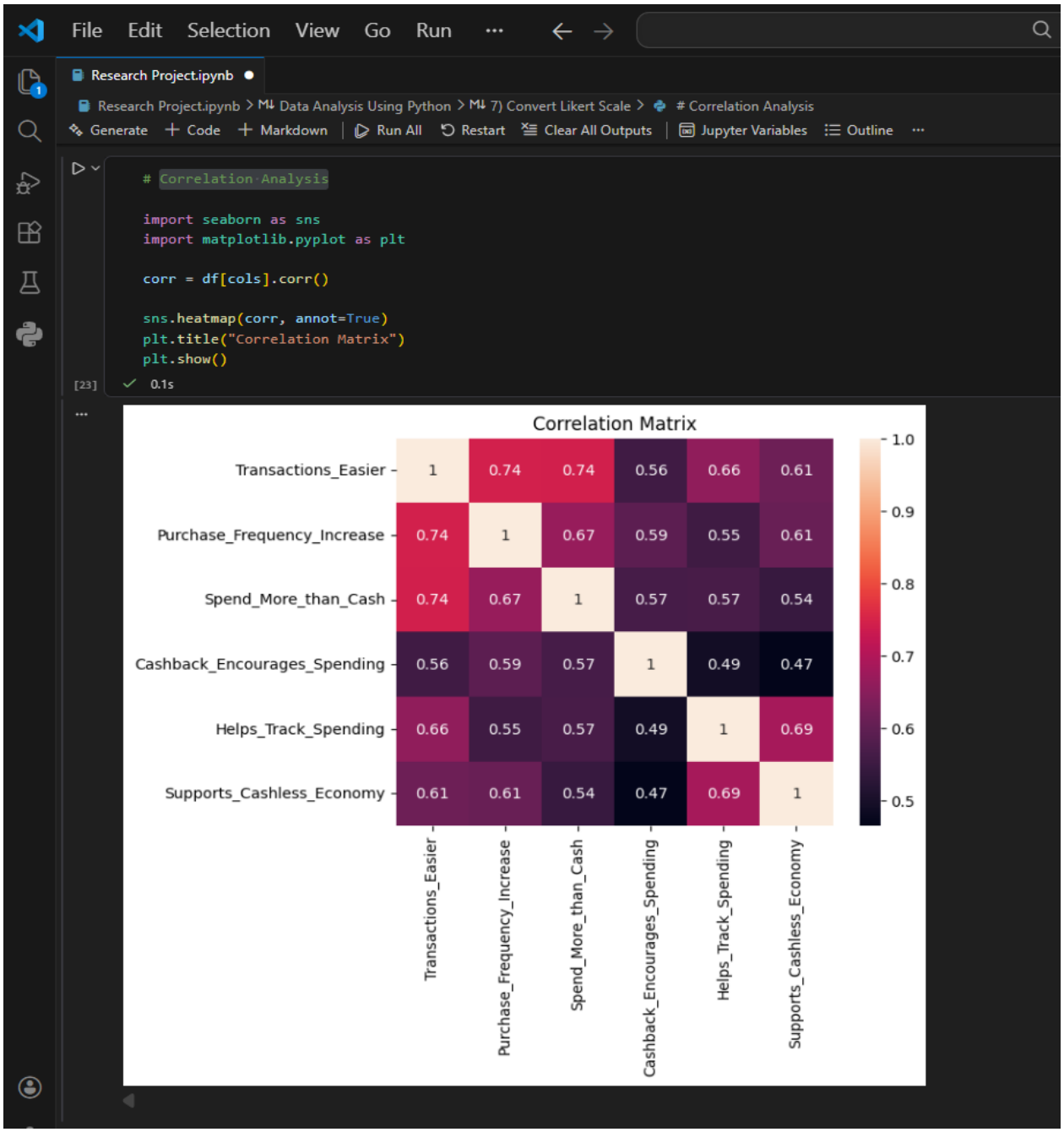


IMAGE NO. 5.19 Correlation Analysis

INTERPRETATION:

The correlation matrix shows a positive relationship between variables related to digital payment usage and consumer spending behaviour. This suggests that increased adoption of digital payments leads to increased spending.

PRICE SEGMENTATION ANALYSIS

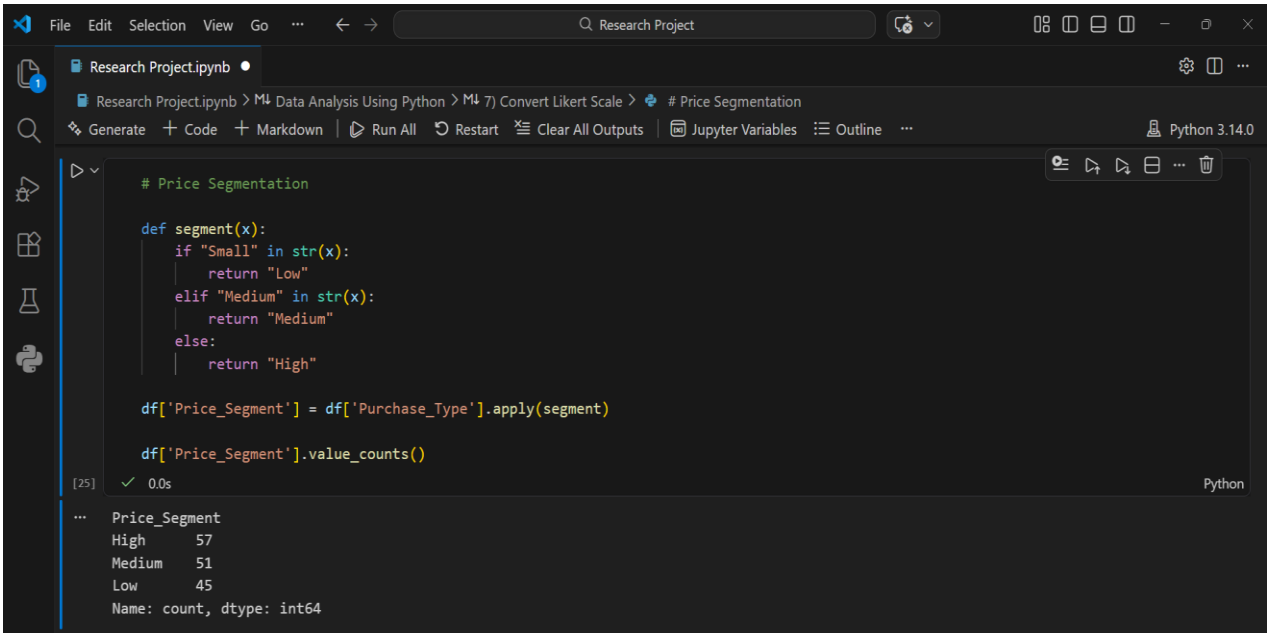


IMAGE NO. 5.20 Price Segmentation Analysis

INTERPRETATION:

Respondents are divided into low, medium and high spending segments based on their purchase behaviour. The majority fall into the medium segment, followed by high and low segments. This classification helps in understanding how different types of consumers behave.

COMPARE SPENDING ANALYSIS

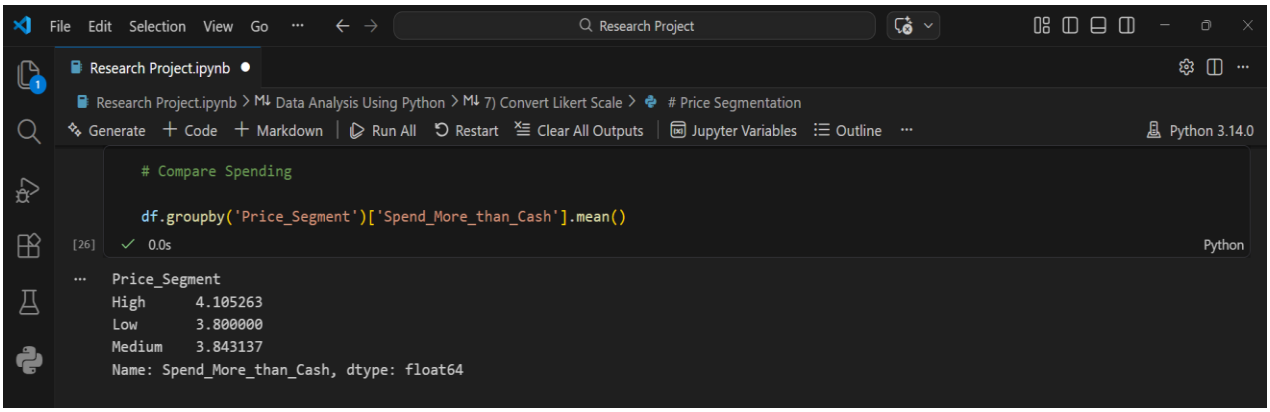


IMAGE NO. 5.21 Compare Spending Analysis

INTERPRETATION:

The comparison shows that spending behaviour varies across different price segments, with medium and high segments showing higher spending levels. This indicates that digital payments influence consumers across all segments.

5.3 HYPOTHESIS TESTING

The Study uses the Chi-Square test to examine whether there is significant association between the variables. Each hypothesis is tested separately and the result are the interpreted in simple and clear manner.

Count of Lickert Scale Questions:

Questions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
12) Do you think digital payments make transactions easier?	12	7	17	42	75
13) Has your frequency of purchases increased after using digital payments?	6	13	20	49	65
14) Do you feel you spend more money when using digital payments compared to cash?	6	19	25	33	70
15) Do cashback offer and discounts encourage you to spend more?	10	20	29	35	59
16) Does digital payment make online shopping more frequent for you?	13	9	36	38	57
23) Digital payment systems help me track and manage my spending.	12	14	23	50	54
24) Digital payments contribute to the development of a cashless economy	9	6	35	38	65

Table No. 5.2 Count of Lickert Scale Questions

Hypothesis 1:

Statement:

H0: Transactions_Easier has no significant impact on **Spend_More_than_Cash**

H1: Transactions_Easier has a significant impact on **Spend_More_than_Cash**

Test Used:

The Chi-Square Test is used to check whether there is relation between Transaction Easier and Spend More Than Cash

Observed Frequency Contingency Tabel:

	Spend More Than Cash					
Transaction Easier ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	Grand Total
Strongly Disagree	4	6	2			12
Disagree	1	5		1		7
Neutral		4	10	2	1	17
Agree	1	2	6	21	12	42
Strongly Agree		2	7	9	57	75
Grand Total	6	19	25	33	70	153

Table No. 5.3 Observed frequency table for Hypothesis 1

Expected Frequency:

$$Expected = \frac{Row\ Total * Column\ Total}{Grand\ Total}$$

	Spend More Than Cash				
Transaction Easier ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree	0.47	1.49	1.96	2.59	5.49
Disagree	0.27	0.87	1.14	1.51	3.20
Neutral	0.67	2.11	2.78	3.67	7.78
Agree	1.65	5.22	6.86	9.06	19.22
Strongly Agree	2.94	9.31	12.25	16.18	34.31

Table No. 5.4 Expected frequency table for Hypothesis 1

Chi Square Value Table:

Formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

	Spend More Than Cash				
Transaction Easier	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree ↓	26.51	13.65	0.00	2.59	5.49
Disagree	1.97	19.61	1.14	0.17	3.20
Neutral	0.67	1.69	18.75	0.76	5.91
Agree	0.26	1.99	0.11	15.74	2.71
Strongly Agree	2.94	5.74	2.25	3.19	15.01

Table No. 5.5 Chi-Square value table for Hypothesis 1

Chi Square Value = 152.04 (sum of all chi-square values)

Degree of Freedom = (r-1) (c-1)

= (5-1) (5-1) = 16

Critical Value = CHISQ.INV.RT(0.05,16)

= 26.296

Decision Rule:

- If Calculated Value > Critical Value = Reject H0
- If Calculated value < Critical value = Accept H0

INTERPRETATION:

The Chi-square statistic was calculated by applying the formula $\Sigma(O-E)^2/E$ for all cells. The calculated value is 152.04 with 16 degrees of freedom. And the Critical value is 26.296. Calculated Value is greater than Critical value. Therefore, the null hypothesis is rejected and it is concluded that there is a significant association between ease of digital payments and consumer spending behaviour.

Hypothesis 2:

Statement:

H0: Purchase_Frequency_Increase has no significant impact on **Spend_More_than_Cash**

H1: Purchase_Frequency_Increase has a significant impact on **Spend_More_than_Cash**

Test Used:

The Chi-Square Test is used to check whether there is relation between Purchase Frequency Increase and Spend More Than Cash

Observed Frequency Contingency Tabel:

	Spend More Than Cash					
Purchase Frequency Increase ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	Grand Total
Strongly Disagree	2	3		1		6
Disagree	1	8	4			13
Neutral	2	3	6	5	4	20
Agree		4	11	19	15	49
Strongly Agree	1	1	4	8	51	65
Grand Total	6	19	25	33	70	153

Table No. 5.6 Observed frequency table for Hypothesis 2

Expected Frequency:

$$Expected = \frac{Row\ Total * Column\ Total}{Grand\ Total}$$

	Spend More Than Cash				
Purchase Frequency Increase ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree	0.24	0.75	0.98	1.29	2.75
Disagree	0.51	1.61	2.12	2.80	5.95
Neutral	0.78	2.48	3.27	4.31	9.15
Agree	1.92	6.08	8.01	10.57	22.42
Strongly Agree	2.55	8.07	10.62	14.02	29.74

Table No. 5.7 Expected frequency table for Hypothesis 2

Chi Square Value Table:

Formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

	Spend More Than Cash				
Purchase Frequency Increase ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree	12.91	6.75	0.98	0.07	2.75
Disagree	0.47	25.36	1.67	2.80	5.95
Neutral	1.91	0.11	6.13	0.11	2.90
Agree	1.92	0.71	1.12	6.72	2.46
Strongly Agree	0.94	6.19	4.13	2.58	15.20

Table No. 5.8 Chi-Square value table for Hypothesis 2

Chi Square Value = 112.83 (sum of all chi-square values)

Degree of Freedom = (r-1) (c-1)

= (5-1) (5-1) = 16

Critical Value = CHISQ.INV.RT(0.05,16)

= 26.296

Decision Rule:

- If Calculated Value > Critical Value = Reject H0
- If Calculated value < Critical value = Accept H0

INTERPRETATION:

The Chi-square statistic was calculated by applying the formula $\Sigma(O-E)^2/E$ for all cells. The calculated value is 112.83 with 16 degrees of freedom. And the Critical value is 26.296. Calculated Value is greater than Critical value. Therefore, the null hypothesis is rejected and it is concluded that there is a significant association between ease of digital payments and consumer spending behaviour.

Hypothesis 3:

Statement:

H0: Cashback_Encourages_Spending has no significant impact on **Spend_More_than_Cash**

H1: Cashback_Encourages_Spending has a significant impact on **Spend_More_than_Cash**

Test Used:

The Chi-Square Test is used to check whether there is relation between Cashback Encourages Spending and Spend More Than Cash

Observed Frequency Contingency Tabel:

	Spend More Than Cash					
Cashback Encourages Spending ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree	Grand Total
Strongly Disagree	1	2	2	2	3	10
Disagree	3	8	4	3	2	20
Neutral	2	7	7	7	6	29
Agree		1	8	16	10	35
Strongly Agree		1	4	5	49	59
Grand Total	6	19	25	33	70	153

Table No. 5.9 Observed frequency table for Hypothesis 3

Expected Value Frequency:

$$Expected = \frac{Row\ Total * Column\ Total}{Grand\ Total}$$

	Spend More Than Cash				
Cashback Encourage Spending ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree	0.39	1.24	1.63	2.16	4.58
Disagree	0.78	2.48	3.27	4.31	9.15
Neutral	1.14	3.60	4.74	6.25	13.27
Agree	1.37	4.35	6.63	7.55	16.01
Strongly Agree	2.31	7.33	9.64	12.73	26.99

Table No. 5.10 Expected frequency table for Hypothesis 3

Chi Square Value Table:

Formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

	Spend More Than Cash				
Cashback Encourage Spending ↓	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Strongly Disagree	0.95	0.47	0.08	0.01	0.55
Disagree	6.32	12.29	0.16	0.40	5.59
Neutral	0.65	3.21	1.08	0.09	3.98
Agree	1.37	2.58	0.28	9.46	2.26
Strongly Agree	2.31	5.47	3.30	4.69	17.95

Table No. 5.11 Chi-Square value table for Hypothesis 3

Chi Square Value = 85.49 (sum of all chi-square values)

Degree of Freedom = (r-1) (c-1)

= (5-1) (5-1) = 16

Critical Value = CHISQ.INV.RT(0.05,16)

= 26.296

Decision Rule:

- If Calculated Value > Critical Value = Reject H0
- If Calculated value < Critical value = Accept H0

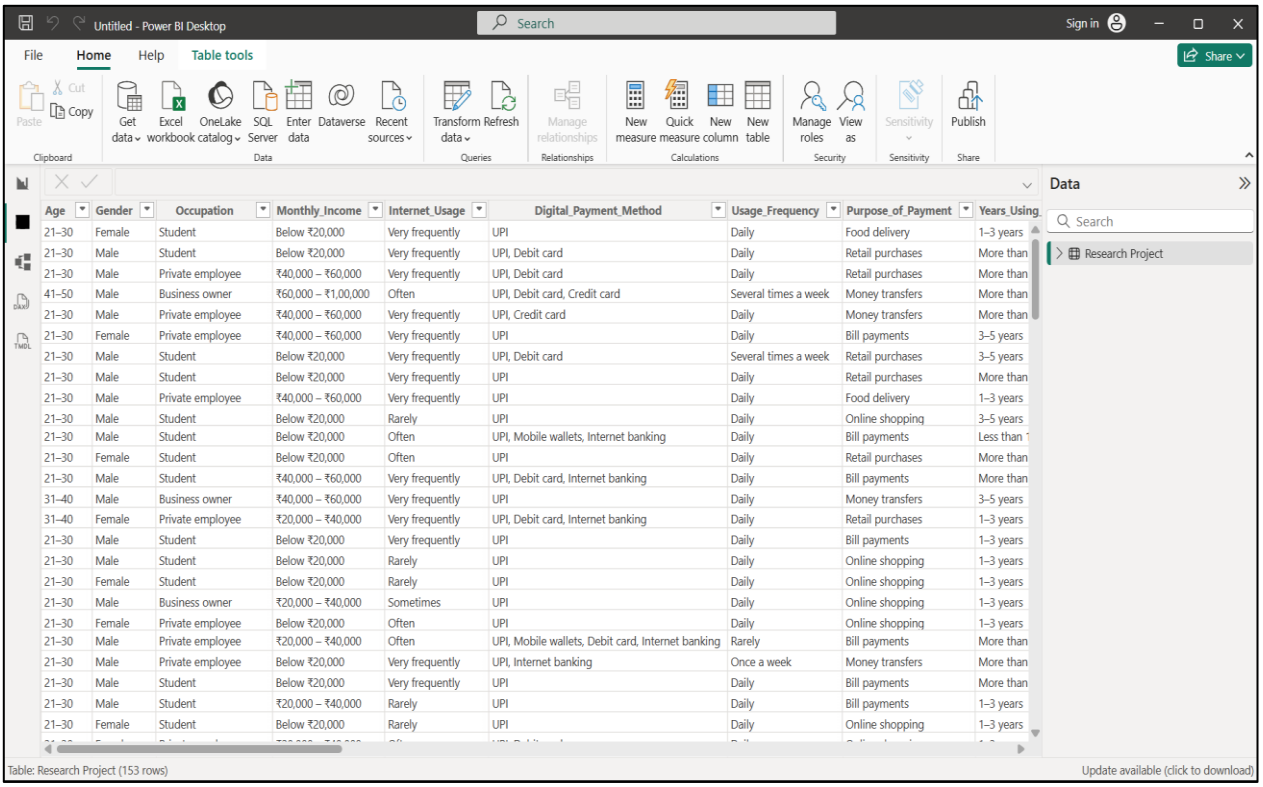
INTERPRETATION:

The Chi-square statistic was calculated by applying the formula $\Sigma(O-E)^2/E$ for all cells. The calculated value is 85.49 with 16 degrees of freedom. And the Critical value is 26.296. Calculated Value is greater than Critical value. Therefore, the null hypothesis is rejected and it is concluded that there is a significant association between ease of digital payments and consumer spending behaviour.

5.4 DATA ANALYSIS THROUGH POWER BI

GET DATA IN POWER BI

Import “Digital Payment Adoption and Consumer Spending Behaviour Survey.xlsx” sheet into Power BI



Age	Gender	Occupation	Monthly_Income	Internet_Usage	Digital_Payment_Method	Usage_Frequency	Purpose_of_Payment	Years_Using
21–30	Female	Student	Below ₹20,000	Very frequently	UPI	Daily	Food delivery	1–3 years
21–30	Male	Student	Below ₹20,000	Very frequently	UPI, Debit card	Daily	Retail purchases	More than
21–30	Male	Private employee	₹40,000 – ₹60,000	Very frequently	UPI, Debit card	Daily	Retail purchases	More than
41–50	Male	Business owner	₹60,000 – ₹1,00,000	Often	UPI, Debit card, Credit card	Several times a week	Money transfers	More than
21–30	Male	Private employee	₹40,000 – ₹60,000	Very frequently	UPI, Credit card	Daily	Money transfers	More than
21–30	Female	Private employee	₹40,000 – ₹60,000	Very frequently	UPI	Daily	Bill payments	3–5 years
21–30	Male	Student	Below ₹20,000	Very frequently	UPI, Debit card	Several times a week	Retail purchases	3–5 years
21–30	Male	Student	Below ₹20,000	Very frequently	UPI	Daily	Retail purchases	More than
21–30	Male	Private employee	₹40,000 – ₹60,000	Very frequently	UPI	Daily	Food delivery	1–3 years
21–30	Male	Student	Below ₹20,000	Rarely	UPI	Daily	Online shopping	3–5 years
21–30	Male	Student	Below ₹20,000	Often	UPI, Mobile wallets, Internet banking	Daily	Bill payments	Less than
21–30	Female	Student	Below ₹20,000	Often	UPI	Daily	Retail purchases	More than
21–30	Male	Student	₹40,000 – ₹60,000	Very frequently	UPI, Debit card, Internet banking	Daily	Bill payments	More than
31–40	Male	Business owner	₹40,000 – ₹60,000	Very frequently	UPI	Daily	Money transfers	3–5 years
31–40	Female	Private employee	₹20,000 – ₹40,000	Very frequently	UPI, Debit card, Internet banking	Daily	Retail purchases	1–3 years
21–30	Male	Student	Below ₹20,000	Very frequently	UPI	Daily	Bill payments	1–3 years
21–30	Male	Student	Below ₹20,000	Rarely	UPI	Daily	Online shopping	1–3 years
21–30	Female	Student	Below ₹20,000	Rarely	UPI	Daily	Online shopping	1–3 years
21–30	Male	Business owner	₹20,000 – ₹40,000	Sometimes	UPI	Daily	Online shopping	1–3 years
21–30	Female	Private employee	Below ₹20,000	Often	UPI	Daily	Online shopping	1–3 years
21–30	Male	Private employee	₹20,000 – ₹40,000	Often	UPI, Mobile wallets, Debit card, Internet banking	Rarely	Bill payments	More than
21–30	Male	Private employee	Below ₹20,000	Very frequently	UPI, Internet banking	Once a week	Money transfers	More than
21–30	Male	Student	Below ₹20,000	Very frequently	UPI	Daily	Bill payments	More than
21–30	Male	Student	₹20,000 – ₹40,000	Rarely	UPI	Daily	Bill payments	1–3 years
21–30	Female	Student	Below ₹20,000	Rarely	UPI	Daily	Online shopping	1–3 years

IMAGE NO. 5.22 Get Data in Power BI

Create Measures (DAX):

1. Avg_Convenience = AVERAGE('Research Project' [Transactions_Easier_Num])
2. Avg_Spending = AVERAGE('Research Project'[Spend_More_than_Cash_Num])

Convert Likert Scale into Numbers:

```
Cashback_Encourages_Spending_Num =
SWITCH(
        'Research Project'[Cashback_Encourages_Spending],
        "Strongly Disagree",1,
        "Disagree",2,
        "Neutral",3,
        "Agree",4,
        "Strongly Agree",5
)
```

DIGITAL PAYMENT ADOPTION & CONSUMER SPENDING BEHAVIOUR ANALYSIS DASHBOARD:

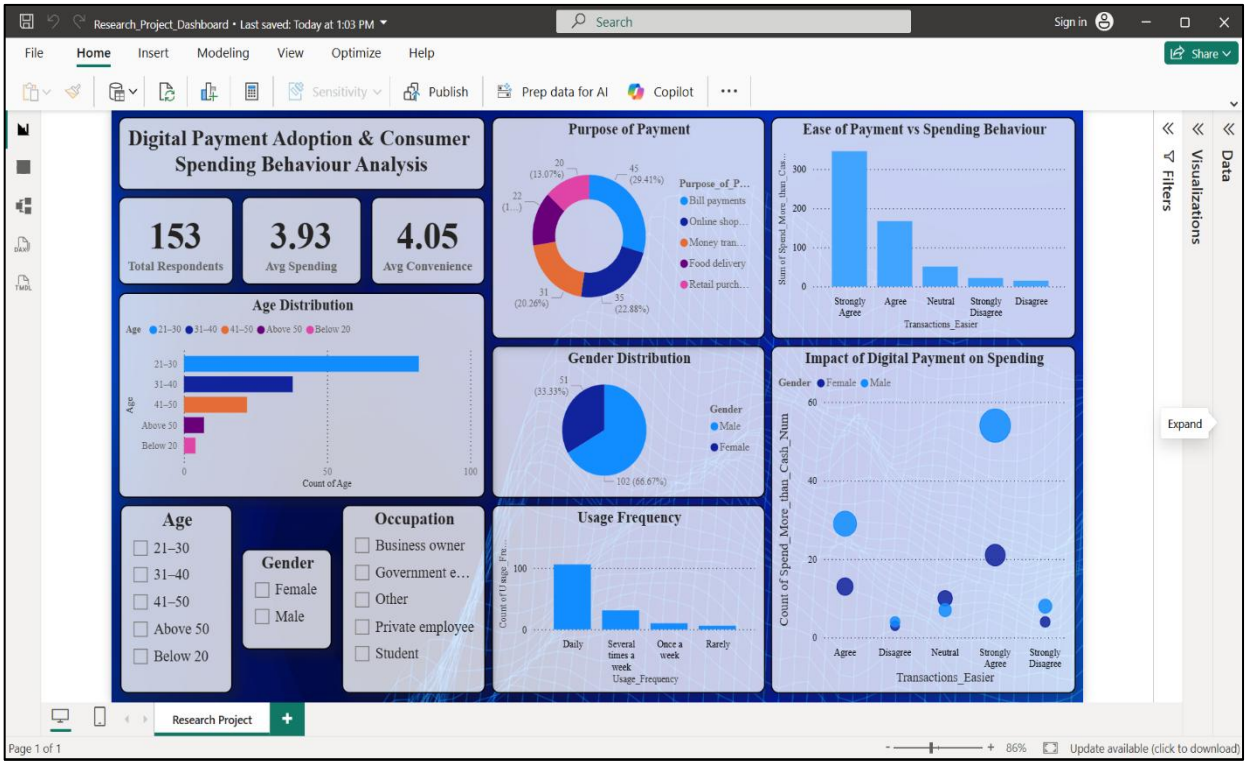


IMAGE NO. 5.23 Digital Payment Adoption & Consumer Spending Behaviour Analysis Dashboard

INTERPRETATION:

The Power BI dashboard created for this study gives a clear and complete picture of how digital payment systems are affecting consumer spending behaviour. It is based on the responses collected from 153 participants and the data has been presented in a simple and visual way so that the insights are easy to understand.

At the top of the dashboard, key highlights are shown. The total number of respondents is 153, which provides a good base for analysis. The average spending score is 3.93, which means that most people agree that they tend to spend more when using digital payments. Along with this, the average convenience score is 4.05, showing that users generally find digital payments easy, fast and comfortable to use in their daily life.

Looking at the demographic details, it is clear that the majority of respondents belong to the 21–30 age group, making young individuals the main users of digital payment systems. The 31–40 age group also shows some participation, while older age groups have comparatively fewer users.

In terms of gender, around 66.67% (102 respondents) are male and 33.33% (51 respondents) are female. Even though male participation is higher, digital payment usage is seen across both genders.

When we look at how frequently people use digital payments, the dashboard shows that most respondents use them on a daily basis, with more than 100 users falling in this category. Some users make payments several times a week, while very few use them occasionally. This clearly shows that digital payments have become a regular part of everyday transactions.

The purpose of using digital payments is also quite clear. Around 29.41% of users mainly use them for bill payments, followed by 22.88% for online shopping and 20.26% for money transfers. The remaining users use digital payments for food delivery and retail purchases. This shows that digital payments are not limited to one activity but are used in multiple areas of daily life.

One of the most important parts of the dashboard is the analysis of spending behaviour. It can be seen that respondents who strongly agree that digital payments are easy to use also show the highest level of spending. Similarly, those who selected “Agree” also show higher spending compared to neutral or disagree groups. This clearly indicates that ease and convenience directly influence how much people spend.

The scatter plot further supports this finding. It shows a positive relationship between ease of transactions and spending behaviour. In simple terms, as people become more comfortable using digital payments, their spending also increases. This pattern is visible across both male and female respondents.

Overall, the dashboard clearly shows that digital payments are widely accepted, especially among young users and are used very frequently. The high convenience score (4.05) and spending score (3.93) suggest that digital payments not only make transactions easier but also encourage people to spend more. Factors like speed, ease of use and cashback offers play an important role in this behaviour.

In conclusion, the dashboard strongly supports the idea that digital payment adoption has a positive impact on consumer spending behaviour. It shows how digital technology is changing the way people make payments and manage their spending in today’s modern economy.

CHAPTER VI

RESULTS AND FINIDINGS

CHAPTER VI

RESULTS AND FINIDINGS

This section presents the key results and findings derived from the data analysis conducted using Python and Power BI. The study aims to understand the impact of digital payment adoption on consumer spending behaviour. The findings are based on responses collected from 153 participants and are supported by statistical analysis and visual representations.

1. Demographic Findings

The analysis of demographic data shows that a majority of respondents belong to the younger age group, particularly between 21–30 years. This indicates that younger individuals are more inclined towards adopting digital payment methods due to their familiarity with technology.

In terms of gender distribution, both male and female respondents have actively participated in digital payment usage. This suggests that digital payment systems are widely accepted across different genders and are not limited to a specific group.

2. Digital Payment Usage Patterns

The findings reveal that digital payment systems are used frequently by the respondents. A significant portion of users reported using digital payments on a daily or weekly basis. This highlights the growing dependence on digital platforms for routine financial transactions.

The primary purposes of using digital payments include online shopping, bill payments and money transfers. This reflects the convenience, speed and accessibility of digital payment systems in everyday life.

3. Perception Towards Digital Payments

The study found that most respondents perceive digital payments as easy to use and convenient. The majority agreed that digital transactions save time and effort compared to traditional cash-based methods.

This positive perception plays a crucial role in increasing the adoption rate of digital payments and encourages users to rely more on these systems.

4. Impact on Consumer Spending Behaviour

One of the key findings of this study is that digital payment adoption has a noticeable impact on consumer spending behaviour.

The analysis shows that respondents who frequently use digital payments tend to spend more compared to those who rely on cash. The ease of making payments, along with quick transaction processes, reduces the psychological barrier associated with spending.

5. Influence of Cashback and Offers

The study highlights that cashback offers, discounts and promotional incentives significantly influence consumer spending.

Many respondents agreed that they are more likely to make purchases when such offers are available. This indicates that marketing strategies linked with digital payments play an important role in increasing consumer expenditure.

6. Relationship Between Convenience and Spending

The statistical and visual analysis (especially scatter plots and correlation analysis) shows a positive relationship between convenience and spending behaviour.

As the ease of using digital payment systems increases, consumers are more likely to engage in higher spending. This supports the idea that convenience is a major driving factor influencing consumer behaviour.

7. Price Segmentation Findings

The study also categorized respondents into different spending segments such as low, medium and high.

The findings suggest that digital payment systems influence all segments, but the effect is slightly stronger among medium and high spending groups. These groups tend to make more frequent and higher-value transactions using digital platforms.

8. Hypothesis Testing Results

Hypothesis	Variables Tested	Chi-Square Value	Critical Value	Result
H1	Transactions_Easier vs. Spend_More_than_Cash	152.04	26.296	Accepted
H2	Purchase_Frequency_Increase vs. Spend_More_than_Cash	112.83	26.296	Accepted
H3	Cashback_Encourages_Spending vs. Spend_More_than_Cash	85.49	26.296	Accepted

Table No. 5.12 Summary of Hypothesis Testing Result

The hypothesis testing conducted using statistical tools Chi-Square Test indicates that there is a significant relationship between digital payment adoption and consumer spending behaviour.

Based on the rules, the null hypothesis (that digital payments have no impact on spending behaviour) is rejected, while the alternative hypothesis is accepted. This confirms that digital payment adoption positively influences consumer spending.

9. Overall Key Findings

Based on the analysis, the following major findings can be summarized:

- Digital payment adoption is high among younger individuals
- Digital payments are used frequently for daily transactions
- Consumers perceive digital payments as convenient and easy to use
- Increased usage of digital payments leads to higher spending
- Cashback and promotional offers encourage additional spending
- There is a positive relationship between convenience and spending behaviour
- Digital payment systems impact all consumer segments

CHAPTER VII

RECOMMENDATIONS FOR FURTHER RESEARCH

CHAPTER VII

RECOMMENDATIONS FOR FURTHER RESEARCH

This study focused on understanding the impact of digital payment adoption on consumer spending behaviour using a limited sample and specific variables. While the findings provide useful insights, there are several areas where further research can be carried out to gain a deeper and broader understanding of this topic.

Firstly, future studies can be conducted with a larger sample size and across different regions or cities. This will help in getting more diverse responses and improving the generalizability of the results. Including rural areas along with urban populations can also provide a better comparison of digital payment adoption patterns.

Secondly, researchers can explore additional factors that may influence consumer spending behaviour. Variables such as financial literacy, trust in digital platforms, security concerns and technological awareness can be included to understand their role in shaping user behaviour.

Another important area for further research is to conduct a comparative study between digital payments and traditional cash transactions. This would help in clearly identifying how consumer behaviour differs between the two modes of payment.

Future research can also focus on specific digital payment platforms such as UPI, mobile wallets or credit/debit cards. Analysing user preferences and behaviour across different platforms can provide more detailed insights.

Longitudinal studies can be conducted to examine how consumer behaviour changes over time with the increasing adoption of digital payments. This will help in understanding long-term trends and the sustainability of digital payment usage.

Additionally, researchers can explore the psychological aspects of spending behaviour, such as impulse buying and emotional spending, in the context of digital payments.

Lastly, advanced analytical techniques such as machine learning and predictive modelling can be used in future studies to predict consumer spending patterns and identify key influencing factors more accurately.

CHAPTER VIII

REFERENCES

CHAPTER VIII

REFERENCES

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ANNEXURE

Annexure A: Questionnaire

Section 1: Demographic Information

1) Name of the Respondent?
2) What is your age group?

a) Below 20
b) 21–30
c) 31–40
d) 41–50
e) Above 50

3) What is your gender?

a) Male
b) Female

4) What is your occupation?

a) Student
b) Private employee
c) Government employee
d) Business owner
e) Other

5) What is your monthly income level?

a) Below ₹20,000
b) ₹20,000 – ₹40,000
c) ₹40,000 – ₹60,000
d) ₹60,000 – ₹1,00,000
e) Above ₹1,00,000

6) How often do you use the internet on your smartphone?

a) Rarely
b) Sometimes
c) Often
d) Very frequently

Section 2: Digital Payment Usage

7) Which digital payment methods do you use?

a) UPI

b) Mobile wallets
c) Debit card
d) Credit card
e) Internet banking

8) How frequently do you use digital payment systems?
a) Daily
b) Several times a week
c) Once a week
d) Rarely

9) For what purposes do you mostly use digital payments?
a) Online shopping
b) Bill payments
c) Food delivery
d) Retail purchases
e) Money transfers

10) How long have you been using digital payment methods?
a) Less than 1 year
b) 1–3 years
c) 3–5 years
d) More than 5 years

11) What is your main reason for using digital payments?
a) Convenience
b) Speed of transactions
c) Cashback/discount offers
d) Safety and security
e) Avoid carrying cash

Section 3: Consumer Spending Behaviour

Scale Guide: 1 = Strongly Disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly Agree

12) Do you think digital payments make transactions easier?

13) Has your frequency of purchases increased after using digital payments?

14) Do you feel you spend more money when using digital payments compared to cash?

15) Do cashback offer and discounts encourage you to spend more?

16) Does digital payment make online shopping more frequent for you?

17) How often do you make impulse purchases using digital payments?

a) Very often
b) Often
c) Sometimes
d) Rarely
e) Never

18) Which type of purchase do you mostly make using digital payments?

a) Small daily purchases
b) Medium purchases
c) Large purchases
d) All types

Section 4: Consumer Perception of Digital Payments

19) How would you rate the convenience of digital payment systems?

a) Very High
b) High
c) Moderate
d) Low

20) How secure do you think digital payment systems are?

a) Very Secure
b) Secure
c) Neutral
d) Not Secure

21) Have you ever experienced fraud or security issues while using digital payments?

a) Yes
b) No

22) Do you trust digital payment platforms for large transactions?

a) Yes
b) No
c) Sometimes

23) Digital payment systems help me track and manage my spending.

a) Strongly disagree

b) Disagree

c) Neutral

d) Agree

e) Strongly agree

24) Digital payments contribute to the development of a cashless economy.

a) Strongly disagree

b) Disagree

c) Neutral

d) Agree

e) Strongly agree

25) Overall, how satisfied are you with digital payment systems?

a) Very satisfied

b) Satisfied

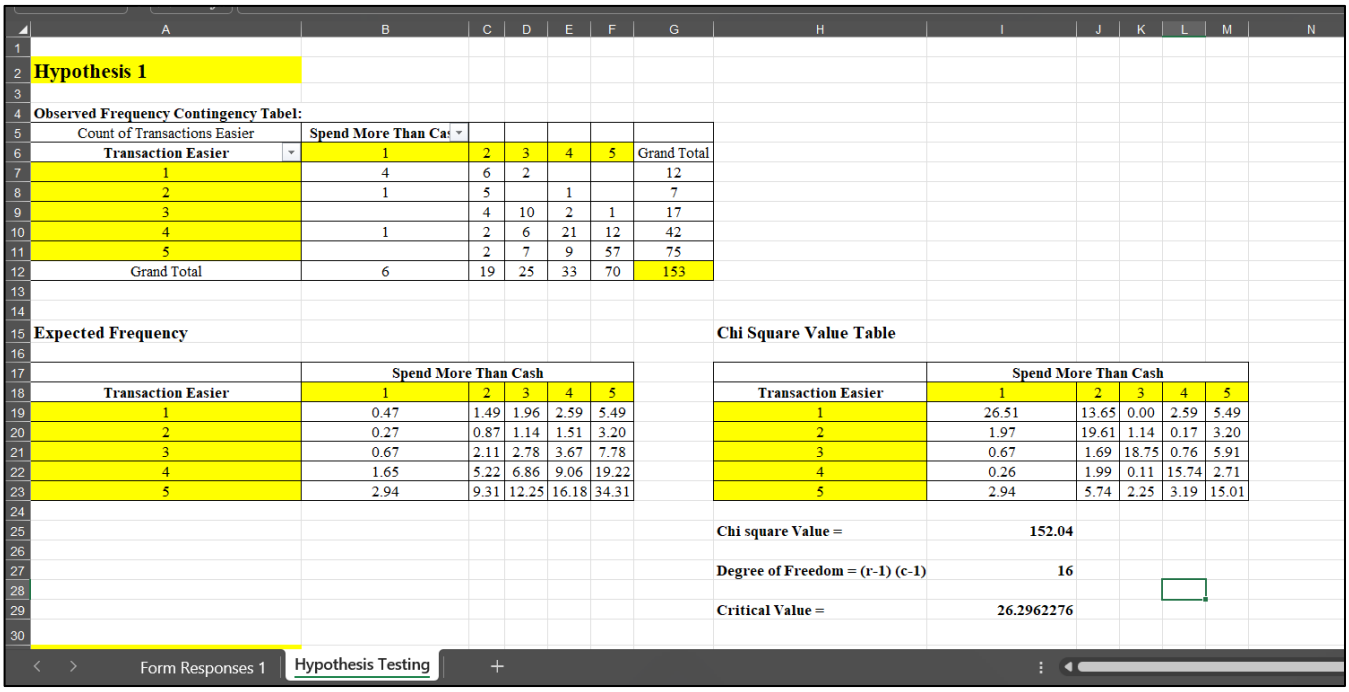
c) Neutral

d) Dissatisfied

e) Very dissatisfied

Annexure B: Chi-Square Calculation Table

Hypothesis 1:



Hypothesis 1

Observed Frequency Contingency Tabel:

Count of Transactions Easier	Spend More Than Ca					
Transaction Easier	1	2	3	4	5	Grand Total
1	4	6	2			12
2	1	5		1		7
3		4	10	2	1	17
4	1	2	6	21	12	42
5		2	7	9	57	75
Grand Total	6	19	25	33	70	153

Expected Frequency

	Spend More Than Cash				
Transaction Easier	1	2	3	4	5
1	0.47	1.49	1.96	2.59	5.49
2	0.27	0.87	1.14	1.51	3.20
3	0.67	2.11	2.78	3.67	7.78
4	1.65	5.22	6.86	9.06	19.22
5	2.94	9.31	12.25	16.18	34.31

Chi Square Value Table

	Spend More Than Cash				
Transaction Easier	1	2	3	4	5
1	26.51	13.65	0.00	2.59	5.49
2	1.97	19.61	1.14	0.17	3.20
3	0.67	1.69	18.75	0.76	5.91
4	0.26	1.99	0.11	15.74	2.71
5	2.94	5.74	2.25	3.19	15.01

Chi square Value = 152.04

Degree of Freedom = (r-1) (c-1) 16

Critical Value = 26.2962276

Form Responses 1 | Hypothesis Testing

Image: Chi-Square Calculation for Hypothesis 1

Hypothesis 2:

Hypothesis 2

Observed Frequency Contingency Tabel:

Count of Purchase_Frequency_Increase	Spend More Than Ca					
Purchaase Frequency Increase	1	2	3	4	5	Grand Total
1	2	3		1		6
2	1	8	4			13
3	2	3	6	5	4	20
4		4	11	19	15	49
5	1	1	4	8	51	65
Grand Total	6	19	25	33	70	153

Expected Frequency Table

	Spend More Than Cash				
Purchase Frequency Increase	1	2	3	4	5
1	0.24	0.75	0.98	1.29	2.75
2	0.51	1.61	2.12	2.80	5.95
3	0.78	2.48	3.27	4.31	9.15
4	1.92	6.08	8.01	10.57	22.42
5	2.55	8.07	10.62	14.02	29.74

Chi Square Value Table

	Spend More Than Cash				
Purchase Frequency Increase	1	2	3	4	5
1	12.91	6.75	0.98	0.07	2.75
2	0.47	25.36	1.67	2.80	5.95
3	1.91	0.11	6.13	0.11	2.90
4	1.92	0.71	1.12	6.72	2.46
5	0.94	6.19	4.13	2.58	15.20

Chi Square value = **112.83**

Degree of Freedom = (r-1)(c-1) **16**

Critical value = **26.30**

Form Responses 1 | Hypothesis Testing

Image: Chi-Square Calculation for Hypothesis 2

Hypothesis 3:

Hypothesis 3

Count of Cashback Encourages Spending	Spend More Than Ca					
Cashback Encourage Spending	1	2	3	4	5	Grand Total
1	1	2	2	2	3	10
2	3	8	4	3	2	20
3	2	7	7	7	6	29
4		1	8	16	10	35
5		1	4	5	49	59
Grand Total	6	19	25	33	70	153

Expected Value Frequency

	Spend More Than Cash				
Cashback Encourage Spending	1	2	3	4	5
1	0.39	1.24	1.63	2.16	4.58
2	0.78	2.48	3.27	4.31	9.15
3	1.14	3.60	4.74	6.25	13.27
4	1.37	4.35	6.63	7.55	16.01
5	2.31	7.33	9.64	12.73	26.99

Chi Square Value Table

	Spend More Than Cash				
Cashback Encourage Spending	1	2	3	4	5
1	0.95	0.47	0.08	0.01	0.55
2	6.32	12.29	0.16	0.40	5.59
3	0.65	3.21	1.08	0.09	3.98
4	1.37	2.58	0.28	9.46	2.26
5	2.31	5.47	3.30	4.69	17.95

Chi Square Value = **85.49**

Degree of Freedom = (r-1)(c-1) **16**

Critical Value = **26.30**

Form Responses 1 | Hypothesis Testing

Image: Chi-Square Calculation for Hypothesis 3

Annexure C: Formulas Used

1. **Expected Frequency:**

$$Expected = \frac{Row\ Total * Column\ Total}{Grand\ Total}$$

2. **Chi Square Value Table:**

   Formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

3. **Degree of Freedom = (r-1) (c-1)**

4. **Critical Value** = CHISQ.INV.RT(probability, deg_freedom)